

Readout/Calibration of Erroneous GKP States

Shraddha Singh

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The Gaussian-Controlled-Rotation (GCR) of arXiv:2504.19992 suppresses the position-uncertainty error of an oscillator-controlled qubit rotation by pairing each position-controlled kick with a compensating momentum-controlled kick. Composed into the Abelian BB1 sequence, it promises a high-fidelity GKP modular readout; here we ask whether that promise survives once the pulses must be physically realizable and the hardware is noisy. The ideal performance for BB1(GCR) was shown through a non-unitary pulse which exactly cancelled the required errors due to Gaussianity. We show, first, that the approximation which this non-unitary pulse mimics cannot be reproduced by the unitary BB1(GCR) pulse, and GCR is the best any deterministic unitary achieves at the no-error point (Sec. 3). Nonetheless, the performance beating or matching GCR at all values of $\langle x \rangle$, be realized by the same pulse *probabilistically* via heralding (Sec. 4): we herald the three BB1 correction pulses, leave the target a deterministic GCR, and post-select on the heralds, so that the $\sim 17\%$ of shots that pass carry the full ideal response yielding a readout infidelity of 5.7×10^{-4} . Additionally, supporting our claim, a variationally optimized BB1(GCR) alternative can yield the same performance, deterministically without any heralding or post-selection (Sec. 6): optimizing the correction amplitudes and axes over the same short conditional-displacement ansatz reaches 1.5×10^{-3} , a factor ~ 5 below bare BB1, and then plateaus with added depth, so no short unitary matches the heralded floor—but the coherent gain BB1(GCR) promises is real and physically achievable either way.

Neither route, however, survives contact with realistic noise. A failed herald ($\sim 83\%$) destroys the logical information through large back-action, and the unknown data state cannot be re-prepared (Sec. 4.6); and once gate *durations* are counted, both the heralded and the compiled deterministic BB1(GCR) share the long 2π correction pulse that a single GCR omits, so decoherence masks their coherent-floor advantage entirely and a single GCR—an order of magnitude shorter, and just as accurate before noise—outperforms them both by $\sim 8\times$ once noise is included (Sec. 6), a verdict a full master-equation simulation confirms (Sec. 7). Two stress tests leave this ranking intact: reading a codeword that idled under decay or dephasing before the read (Sec. 8), and reading one actively stabilized by noisy small-big-small rounds (Sec. 9). BB1(GCR), whether heralded or deterministically compiled, thus emerges not as an end-of-line readout but as a proof that the ideal correction is physically recoverable, and as the tool of choice wherever post-selection or re-preparation is free, as in calibration and known-state preparation (Sec. 5). Unless noted otherwise, all numbers are QuTiP simulations at $\Delta = 0.34$, $|\alpha| = \sqrt{\pi}/2$, and target angle $\theta_t = \pi/2$, with readout operating points at $\langle \hat{x} \rangle \in \sqrt{\pi}\mathbb{Z}$.

1 Setup

We build the readout by replacing each rotation of the Abelian BB1 sequence with a Gaussian-Controlled-Rotation (GCR). We use the *reversed* BB1 ordering, so that each pre-correction acts

on the state left by the preceding ideal rotation—the ordering the GCR pre-correction structure requires,

$$\text{BB1}(\text{GCR}) = \text{GCR}_{\phi_0}\left(\frac{\theta_t}{|\alpha|}\hat{x}\right) \text{GCR}_{\phi_1}\left(\frac{\pi}{|\alpha|}\hat{x}\right) \text{GCR}_{3\phi_1}\left(\frac{2\pi}{|\alpha|}\hat{x}\right) \text{GCR}_{\phi_1}\left(\frac{\pi}{|\alpha|}\hat{x}\right), \quad \phi_1 = \arccos\left(-\frac{\theta_t}{4\pi}\right). \quad (1)$$

whose four constituent rotation angles are $\{\pi, 2\pi, \pi, \theta_t\}$. Each GCR unit is a position kick followed by a momentum pre-correction (its “second pulse”),

$$\text{GCR}_{\phi}\left(\frac{\theta}{|\alpha|}\hat{x}\right) = \underbrace{e^{-i\frac{\theta}{2|\alpha|}\hat{x}\sigma_{\phi}}}_{\text{position kick}} \underbrace{e^{-i\frac{\theta\Delta^2}{2|\alpha|}\hat{p}\sigma_{\gamma}}}_{\text{pre-correction}}, \quad (2)$$

whose pre-correction axis σ_{γ} — the second, momentum-controlled kick of each unit — is fixed by the ideal (error-free) qubit state $|\zeta\rangle$ reached after the preceding rotation, through

$$\sigma_{\gamma} |\zeta\rangle = i \sigma_{\phi} |\zeta\rangle. \quad (3)$$

This condition is what lets the momentum kick undo, in the position basis, the Gaussian position-uncertainty error of the next pulse, exactly as in the GCR construction. The task of the composite pulse is to extract bitwise modular position information: a good readout produces a response $P(+1)$ versus $\langle\hat{x}\rangle$ that approximates a square wave, reading $+1$ where $\lfloor\langle\hat{x}\rangle/(2|\alpha|)\rfloor$ is even and -1 where it is odd. The whole question is how faithfully the pre-correction (2)–(3) can be built.

2 The ideal readout is non-unitary

Equation (3) is solved *exactly*, as an operator, by the anti-Hermitian choice $\sigma_{\gamma} \rightarrow i\sigma_{\phi}$. The pre-correction then becomes

$$e^{-i\frac{\theta\Delta^2}{2|\alpha|}\hat{p}(i\sigma_{\phi})} = e^{+\frac{\theta\Delta^2}{2|\alpha|}\hat{p}\sigma_{\phi}} \equiv M, \quad (4)$$

an “imaginary conditional displacement”: a *non-unitary* reweighting of the Gaussian along $\hat{x}$ that cancels the position-uncertainty error at every angle, not just to leading order. Used as the pre-correction of every GCR unit, it gives an almost perfect square wave (red dashed curve in Fig. 2a), with operating-point error $P(-1)|_{x=0} \approx 4 \times 10^{-7}$. The catch is that $i\sigma_{\phi}$ is anti-Hermitian, so M is not a physical (unitary) gate.

3 Why the physical (unitary) pulse fails — and no unitary variant rescues it

A physical pre-correction must use a Hermitian axis, and this imposes two obstructions. The first is *realizability*: a Hermitian σ_{γ} obeying (3) exists only when the qubit is perpendicular to the pulse axis, $\langle\zeta|\sigma_{\phi}|\zeta\rangle = 0$ (otherwise (3) would equate a real number to an imaginary one), in which case $\sigma_{\gamma} = \hat{n} \times \hat{\phi}$ with $\hat{n}$ the Bloch vector of $|\zeta\rangle$. BB1 does satisfy this at every step, since its π and 2π correction pulses shuttle the qubit between the poles $\pm\hat{z}$ and so keep it perpendicular to the equatorial axes σ_{ϕ} , deferring the sign-dependent readout rotation to the final target pulse. The second obstruction is more serious: even where $\sigma_{\gamma} = \hat{n} \times \hat{\phi}$ exists, the *unitary* pre-correction reproduces the ideal M only to leading order in the pulse angle θ . Because BB1’s 2π pulse is four times the target angle, its higher-order residuals dominate, and there the unitary correction does more harm than applying none at all: the readout washes out entirely (Table 1, Fig. 2a, green curve).

scheme (at $\langle x \rangle = 0$)	$P(-1)$
ideal, non-unitary ($\sigma_\gamma = i\sigma_\phi$)	4.1×10^{-7}
bare BB1 (no GCR)	6.1×10^{-3}
physical (unitary) BB1(GCR) ($\sigma_\gamma = \hat{n} \times \hat{\phi}$)	0.31

Table 1: Operating-point readout error. Only the non-unitary ideal beats bare; the physical *unitary* BB1(GCR) is washed out (0.31, i.e. essentially random, far worse than bare), and no legal unitary variant (the whole sequence, or the $2\pi \rightarrow \pi + \pi$ split) does better.

Nor can splitting the pulses rescue the deterministic readout. The readout peaks sit at $x = 2m|\alpha|$, where a pulse of angle-label η rotates the qubit by $2\eta m$, so a BB1 correction acts as an identity at every peak — fixing one consistent, x -agnostic axis $\sigma_\gamma = \hat{n} \times \hat{\phi}$ — only for $\eta \in \pi\mathbb{Z}$. The only legal unitary freedoms are therefore the whole sequence or the split $2\pi \rightarrow \pi + \pi$; the latter composes to *exactly* the same operator, while any sub- π piece sends different peaks to different qubit states and breaks the axis condition outright. Since a product of real conditional displacements can never reproduce the non-unitary $i\sigma_\phi$ of (4), bare BB1 is the best any unitary sequence achieves (Table 1). The failure is *structural* — the unitary correction is simply the wrong kind of operator — and no tuning of sub-rotations or amplitudes can cure it.

4 The flagged readout: a post-selected modular readout

The only operator that beats bare is the non-unitary $M = e^{c\hat{p}\sigma_\phi}$ of (4) (with $c = \theta\Delta^2/2|\alpha|$), which cannot be a gate. It can be realized *probabilistically* by heralding, giving a *post-selected* readout: the shots in which the heralds succeed carry the full BB1(GCR) response. Heralding only the three *corrections* (not the target) both raises the success probability and keeps the outcome-defining pulse out of the post-selection. This is the flagged readout, the practical payoff of this note.

4.1 Construction

BB1(GCR) has four pulses: three *corrections* $\{\pi, 2\pi, \pi\}$ and a final *target* $\{\pi/2\}$ whose measurement produces the ± 1 bit. We herald the ideal non-unitary M on the three correction pulses only (each with its own ancilla), and leave the target a *deterministic* (unitary) GCR. On the $\sim 17\%$ of shots where all three heralds succeed, the corrections are the exact M and the whole sequence is the ideal BB1(GCR); **these are the shots we keep**.

This is genuine *post-selection*, not a confidence tag on an otherwise-good bit. A failed herald applies the complementary non-unitary Kraus operator, which corrupts the *shared readout qubit*: a failed shot then reads the *wrong* bit ($P(-1)|_{x=0} = 0.58$, worse than a coin flip), and taking every shot regardless of the flag washes the readout out to $P(-1)|_{x=0} \approx 0.26$ — worse than bare BB1. Failed shots must therefore be discarded. Leaving the *target* deterministic (not heralding it too) is what keeps the success probability at $\sim 17\%$ rather than $\sim 7\%$, and keeps the bit-defining pulse out of the post-selection — but it does *not* make the readout deterministic.

4.2 The herald: block-encoding and gate sequence

Heralding realizes M by *block-encoding*. Any M with $\|M\| \leq \lambda$ embeds in a unitary V on system+ancilla with $\langle 0|V|0 \rangle_{\text{anc}} = M/\lambda$, so measuring the ancilla and keeping $|0\rangle$ applies M (up to

normalization) with probability

$$P_{\text{succ}} = \frac{\|M|\psi\rangle\|^2}{\lambda^2}. \quad (5)$$

The herald is a measurement of a *separate ancilla*, not of the readout qubit: projecting the readout qubit would delete the $|e\rangle$ branch that carries the answer, whereas M must *reweight* the oscillator while keeping both qubit branches. Conditioning on the ancilla outcome is exactly what “apply M and renormalize” means in the simulation. For one correction pulse (angle η , axis ϕ , strength $c = \eta\Delta^2/2|\alpha|$) the elementary gadget (Fig. 1) prepares a fresh ancilla A in $|0\rangle$, applies $W = \exp(-ic\hat{p}\sigma_\phi^{(S)} \otimes \sigma_y)$ — the usual momentum-kick conditional displacement (controlled by the system qubit through σ_ϕ), now *additionally* controlled by A through σ_y — and then applies a Hadamard to A and measures it, heralding success on $A = 0$ and discarding $A = 1$. Writing $\Theta = c\hat{p}\sigma_\phi$, the W step gives $W|\psi\rangle|0\rangle = \cos\Theta|\psi\rangle|0\rangle + \sin\Theta|\psi\rangle|1\rangle$, and the Hadamard and measurement project the $A = 0$ branch onto $\frac{1}{\sqrt{2}}(\cos\Theta + \sin\Theta)|\psi\rangle$, while $A = 1$ gives the complementary $e^{-c\hat{p}\sigma_\phi}$.

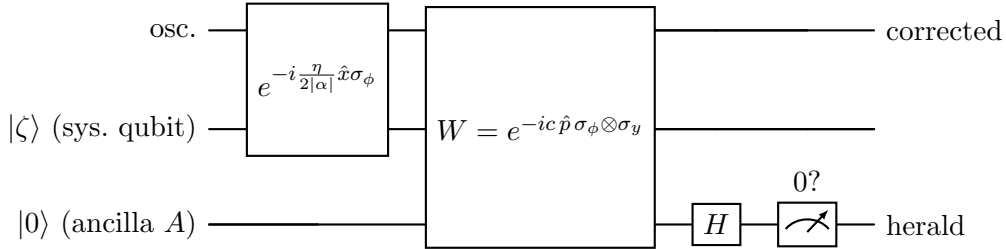


Figure 1: One heralded GCR correction (pulse angle η , axis ϕ , strength $c = \eta\Delta^2/2|\alpha|$). The position kick (first gate) is the ordinary conditional displacement on oscillator+system qubit. The pre-correction is the *ancilla-controlled* conditional displacement $W = e^{-ic\hat{p}\sigma_\phi \otimes \sigma_y}$, followed by a Hadamard on the ancilla and a measurement of the *ancilla* (not the readout qubit). Outcome $|0\rangle$ heralds the non-unitary correction M/λ ; $|1\rangle$ is discarded.

This elementary gadget reproduces M only to *first order* in c , since $\cos\Theta + \sin\Theta \approx 1 + c\hat{p}\sigma_\phi \approx e^{c\hat{p}\sigma_\phi}$. Its success state overlaps the exact $M|\psi\rangle$ with a fidelity that tracks the correction strength $c = \eta\Delta^2/2|\alpha|$: it is 0.996 for the $\pi/2$ target ($c \approx 0.10$) and 0.94 for the π corrections ($c \approx 0.20$), but only 0.47 for the strong 2π correction ($c \approx 0.41$). The 5.7×10^{-4} readout quoted below therefore assumes the *exact* block-encoding of M — as realized in simulation by applying M and renormalizing, and physically by an exact block-encoding of M on the low-energy subspace (QSVT / qubitization) or by iterating the gadget. The single W +Hadamard step is the leading-order intuition, not a faithful implementation of the 2π correction. W itself compiles from the echoed conditional-displacement primitive used for the two-mode gates: a $\text{CD}(c, \sigma_\phi)$ echoed by an ancilla π -pulse toggles the displacement sign with the ancilla.

It is natural to ask whether amplitude amplification could remove the post-selection, since in quantum algorithms one rarely discards the failure branch of a block-encoding: the linear-combination-of-unitaries (LCU) construction is de-randomized by (*oblivious*) *amplitude amplification*, which rotates the state into the success branch with $O(\lambda)$ Grover-like reflections and *no* measurement — this is why LCU-based Hamiltonian simulation is deterministic. That works, however, only when the encoded target is (proportional to) a *unitary*: then $\|M|\psi\rangle\|$ is input-independent and the success/failure pair spans a fixed two-dimensional subspace the reflections close perfectly. Our $M = e^{c\hat{p}\sigma_\phi}$ is a genuine *contraction* — it changes norm, which is the entire source of its power — so oblivious amplification does not apply: the success probability is bounded by $\|M|\psi\rangle\|^2/\|\psi\|^2$ and can be pushed *toward*, but never *to*, unity. The post-selection is therefore intrinsic, not an

artifact of the elementary gadget. Put differently, amplitude amplification de-randomizes exactly the block-encodings whose target could have been a gate in the first place (unitary); M can never join that class, which is the same non-unitarity that made every deterministic attempt above fail.

4.3 Why the target is left deterministic

Heralding *all four* pulses — the target included — reaches the ideal response (4×10^{-7}) on its success branch (Table 2), at $\sim 7\%$ yield. Each herald measures a correction’s *ancilla*, not the readout qubit, so the g/e outcome is preserved: this is a legitimate (if low-yield) heralded readout, not a post-selection on the bit. We nonetheless leave the target a deterministic GCR, which raises the yield to $\sim 17\%$ and uses one fewer ancilla, at the cost of the target’s own χ^4 residual (a floor of 5.7×10^{-4} rather than 4×10^{-7}). Two corollaries follow: the target *must* be a GCR — a bare final rotation gives 4.3×10^{-2} , worse than bare BB1 — and the three corrections are best heralded as a single block (splitting the 2π into $\pi + \pi$ gives the same operator, and post-selecting each pulse separately only lowers the yield).

strength s	0 (bare)	0.25	0.5	0.75	1.0 (ideal)
$P(-1) _{x=0}$	6.1×10^{-3}	2.5×10^{-3}	7.1×10^{-4}	6.6×10^{-5}	4.1×10^{-7}
P_{succ} (global)	1.0	0.45	0.13	7.9×10^{-3}	1.7×10^{-4}
P_{succ} (subspace $n \leq 40$)	1.0	0.62	0.36	0.18	0.071

Table 2: *Aside — four-pulse heralding* (all four corrections heralded): success-branch readout error and success probability vs correction strength s , for the naive global block-encoding and the subspace-tailored one. The subspace bound reaches the ideal (4×10^{-7}) at $\sim 7\%$ yield, and a tenfold improvement over bare at $\sim 36\%$. Since the heralds act on ancillas (not the readout qubit) this is a valid heralded readout; the three-herald readout of the main text is preferred for its higher yield ($\sim 17\%$). Not plotted.

The yield is set by the subspace bound. The naive $\lambda = \|M\|$ over the whole truncated space is wasteful, because $i\sigma_\phi$ amplifies the *unpopulated* high-momentum tail. Restricting the block-encoding to the low-energy input subspace ($n \leq 40$ photons, containing 99.998% of the physical input) gives a far smaller λ and hence a much higher yield — this is what raises the three-herald success from a few percent to $\sim 17\%$.

4.4 Results

On its heralded branch the readout is a correct square wave at every peak (Fig. 2a), with operating-point error $P(-1)|_{x=0} \approx 5.7 \times 10^{-4}$ at herald success $\approx 17\%$; the remaining $\sim 83\%$ are discarded (Sec. 4.6). This floor is set by the *un-heralded* target GCR’s own χ^4 residual, not the full ideal 4×10^{-7} ; the three corrections are otherwise exact. Two comparisons matter, and both use the deterministic finite-energy single-GCR readout as the benchmark.

On the readout error itself (Fig. 2b), the heralding buys *nothing* at a peak: a single deterministic GCR already reaches 5.7×10^{-4} there — both are floored by the same target-GCR residual — and the two curves meet. What the post-selection earns instead is *robustness across the correctable region*. As one approaches the neighbouring peak, where the correct bit flips to -1 so that $P(+1)$ is now the error, the heralded BB1(GCR) response stays sharpest, driving $P(+1)$ down to $\approx 5 \times 10^{-3}$ just off the peak against $\approx 5 \times 10^{-2}$ for the single GCR, which rolls off soonest (the unphysical

ideal reaches $\sim 4 \times 10^{-7}$). This peak-location robustness — BB1’s defining role — is the sole readout-error advantage the post-selection wins over a deterministic GCR.

The back-action tells a similar story (Fig. 2c). On the success branch the readout is nearly QND: the oscillator returns to the input at infidelity $\approx 4 \times 10^{-3}$ across the whole correctable region, whereas bare BB1 mangles an off-peak state, its infidelity peaking at ≈ 0.12 at the bit-flip boundary. This gentleness is again shared with the single GCR ($\approx 4 \times 10^{-3}$) — a virtue of the GCR correction rather than of BB1(GCR) specifically — while the unphysical ideal is gentlest of all ($\approx 6 \times 10^{-7}$, essentially perfect QND). Either way, the 17% of good shots read the bit *and* preserve the state.

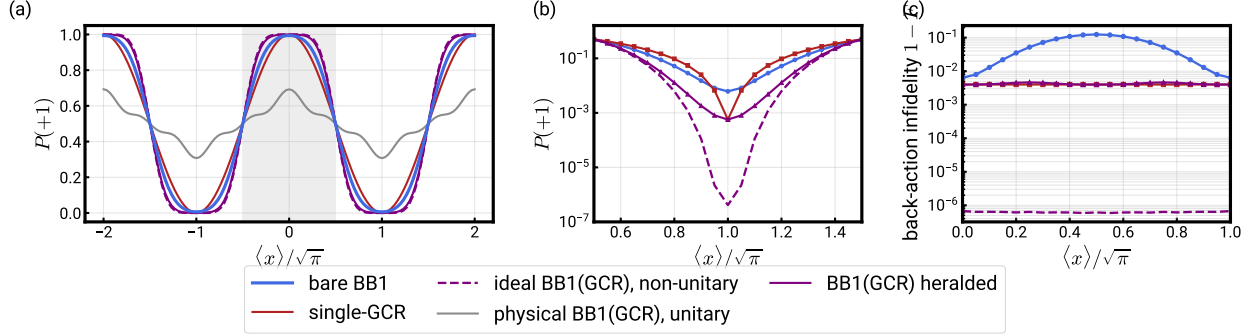


Figure 2: **(a)** full-range readout $P(+1)$: bare BB1 (royalblue), single-GCR (firebrick), the ideal non-unitary BB1(GCR) (purple dashed), the physically-realizable *unitary* BB1(GCR) (grey; $\sigma_\gamma = \hat{n} \times \hat{\phi}$, washes out to ≈ 0.31), and the heralded branch (purple solid, the $\sim 17\%$ -herald branch: herald 3 corrections + deterministic GCR target). **(b)** $\log P(+1)$ zoomed near the neighbouring (odd) peak at $\langle x \rangle / \sqrt{\pi} = 1$, where the correct bit is -1 so $P(+1)$ is the readout error: the ideal reaches $\sim 4 \times 10^{-7}$; among deployable curves heralded BB1(GCR) is sharpest ($\approx 5 \times 10^{-3}$ just off the peak vs single-GCR’s $\approx 5 \times 10^{-2}$), the two meeting the common 5.7×10^{-4} target-GCR floor at the peak. **(c)** oscillator back-action infidelity $1 - F$ (reduced-oscillator state vs the input) over $\langle x \rangle / \sqrt{\pi} \in [0, 1]$ (log): the ideal non-unitary BB1(GCR) is nearly perfectly QND ($\approx 6 \times 10^{-7}$); single-GCR and heralded stay $\approx 4 \times 10^{-3}$; bare BB1 disturbs the state most, peaking at ≈ 0.12 at the bit-flip boundary.

4.5 BB1(GCR) against all the readout schemes

Figure 3 places BB1(GCR) alongside the other GKP readouts — infinite-energy (single CD), finite-energy GCR, Abelian BB1, the deployed GCR-BB1, and its two *block* variants (block A, block B) — on the readout-error metric of the main text. The *ideal* BB1(GCR) (the exact non-unitary correction) drives the error *below the Helstrom bound*, the fundamental limit on any *deterministic* measurement. This is not a contradiction: the ideal is a non-unitary limit, realized in the simulation only by the per-pulse renormalization, and only a non-deterministic (post-selecting) scheme can beat Helstrom.

The catch is the herald success probability. To each non-unitary correction we attach the fraction that the per-pulse `.unit()` renormalization silently keeps, $P_{\text{succ}} = \|M|\psi\rangle\|^2 / \lambda^2$, with λ the largest singular value of the correction over the populated low-energy subspace ($n \leq 40$). For the *heralded* readout (herald the three corrections, deterministic target) this is $\sim 17\%$ (heralding the target too would reach the ideal but at only $\sim 7\%$). Because each herald acts on an ancilla, not the readout qubit, the g/e outcome is preserved — but $\gtrsim 83\%$ of shots are still discarded,

which is why BB1(GCR) is not a usable readout as it stands. But the prize is plain: BB1(GCR) is, in principle, the *best* readout available — below every deterministic scheme and even below Helstrom — so if a route to a higher herald pass rate is found, BB1(GCR) is the readout one should use. Panel (b) is the mirror image for the *logical-1* codeword: the same readout error, now $P(g|\epsilon) = 1 - P(e|\epsilon)$, over the logical-1 peak. It dips to the same floors at the logical-1 peak $\epsilon = \sqrt{\pi}/2$ (ideal $\sim 10^{-10}$, heralded $\sim 5.7 \times 10^{-4}$), confirming the readout is symmetric between the two codewords.

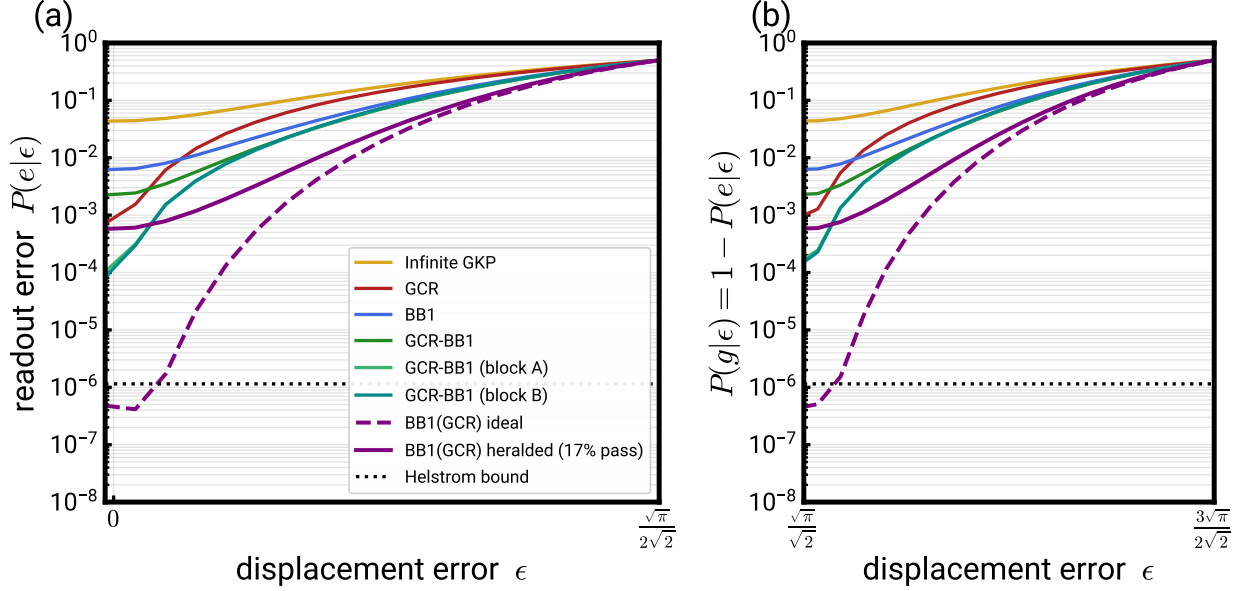


Figure 3: **(a)** Readout error $P(e|\epsilon)$ vs displacement error ϵ for all GKP readouts: infinite-energy (goldenrod), finite-energy GCR (firebrick), Abelian BB1 (royalblue), GCR-BB1 (forestgreen), its two block variants (block A, mediumseagreen; block B, darkcyan), and BB1(GCR) — the non-unitary *ideal* (purple dashed) and the *heralded* readout (purple solid, $\sim 17\%$ herald pass). Dotted: the Helstrom bound ($\approx 1.1 \times 10^{-6}$, computed from the two logical-codeword overlap at $\Delta = 0.34$). The ideal falls below Helstrom because it is a non-unitary limit, not a deterministic measurement. **(b)** the *logical-1* readout error $P(g|\epsilon) = 1 - P(e|\epsilon)$ (log, same curves) over the logical-1 peak: it mirrors (a), dipping to the floor at $\epsilon = \sqrt{\pi}/2$ (ideal $\sim 10^{-10}$, heralded $\sim 5.7 \times 10^{-4}$); the Helstrom bound (dotted) again lies below only the ideal. The displacement error is in units of $\sqrt{\pi}/2\sqrt{2}$ (ticks: panel (a) $\{0, \sqrt{\pi}/2\sqrt{2}\}$; panel (b) $\{\sqrt{\pi}/\sqrt{2}, 3\sqrt{\pi}/2\sqrt{2}\}$).

4.6 On a failed herald: restart, reuse, and back-action

This is an *end-of-line* readout, so the oscillator is discarded afterwards. The back-action is sharply *conditional*: on a successful herald the completed readout is nearly QND — the oscillator returns to the input at fidelity 0.996 (purity 0.999), so the $\sim 17\%$ good shots preserve the state as well as reading the bit — whereas the $\sim 83\%$ failure branch destroys it (oscillator fidelity ≈ 0.24 , though the mean position $\langle \hat{x} \rangle$ is still preserved). So the problem is not back-action per se, but that the destructive branch dominates. When a herald fails, can one recover the sharp read? There are only two ways to try, and neither works. The first would be to start again from the original oscillator state, but this is impossible: the oscillator *is* the unknown data being read, and there is no copy to re-prepare. (This is precisely why the same heralding works cleanly in state

preparation or calibration, where the input is known and can simply be re-prepared, but not for a data readout.) The second would be to reuse the disturbed oscillator with a fresh qubit. Here the coarse bit survives but the precision does not: a failed correction herald applies the complementary non-unitary Kraus operator, which reweights the wavefunction while leaving the mean position $\langle \hat{x} \rangle$ *unchanged* at every peak, so the correct *sign* is still read. The *sharpness*, however, is lost — even a subsequently successful restart gives only $P(-1)|_{x=0} \approx 9 \times 10^{-2}$ after one failed round, already *worse* than the single bare BB1 (6×10^{-3}) that BB1(GCR) was meant to improve, degrading to ~ 0.15 and 0.21 after two and three. The non-unitary shear accumulates and re-correction does not undo it; on the all-fail branch the oscillator’s fidelity to the input is ≈ 0.24 , and no displacement or momentum boost recovers it.

The sharp 5.7×10^{-4} read is thus a *first-attempt* prize available only on the $\sim 17\%$ success branch, and because a failed attempt’s back-action is destructive this is not a scheme that can be “repeated to increase fidelity” — that recourse applies only to readouts whose back-action is small on *every* outcome, such as single GCR or GCR-BB1. The practical verdict is therefore negative: every use carries an $\sim 83\%$ chance of leaving the oscillator in a state that reads *worse than bare BB1* and cannot be re-prepared, so the scheme destroys the logical information more often than it sharpens it. BB1(GCR) is a proof that the non-unitary correction is physically recoverable in post-selectable settings, not a deployable readout.

5 Calibration: realizing the ideal correction, and its noise budget

The $\gtrsim 83\%$ discard that rules BB1(GCR) out as a data readout is a property of reading *unknown* data, whose state cannot be re-prepared or copied. In *calibration* — where the readout is characterized on a *known*, re-preparable GKP state $|\psi\rangle$ — those restrictions lift, and the ideal (non-unitary) correction can be applied *deterministically in expectation*. We give the explicit protocol and, crucially, its noise budget: under realistic decoherence the two natural routes differ by more than an order of magnitude, and the shot-optimal one is the wrong choice.

It helps to recognize what the correction is: each GCR correction $M = e^{c\hat{p}\sigma_\phi} = e^{-\tau H_{\text{eff}}}$ is a small imaginary-time step, with $H_{\text{eff}} = -\hat{p}\sigma_\phi$ a low-weight “Hamiltonian” and $\tau = c = \eta\Delta^2/2|\alpha| \leq 0.41$, so any imaginary-time-evolution method applies. The noise-sensible route is simply repeat-until-success. Because $|\psi\rangle$ is re-preparable, the deterministic-in-expectation realization needs *no* amplitude amplification: we prepare $|\psi\rangle$, run the *shallow* heralded BB1(GCR) of Sec. 4 (one composite pulse, eight conditional displacements, $\sim 6\mu\text{s}$), and keep the shot if the heralds fire; otherwise re-prepare and retry. Heralding three corrections passes with $P_{\text{succ}} \approx 17\%$ (~ 6 attempts; floor 5.7×10^{-4}), and heralding all four returns the ideal response (4×10^{-7}) at $\approx 7\%$ (~ 14 attempts). This pass rate is essentially *noise-independent*: tracking the un-normalized norm against the subspace bound ($\lambda \approx 7.2$ on $n \leq 40$) gives $17.3\% \rightarrow 17.6\%$ under the system channels — decoherence reshapes the state toward larger $\|M|\psi\rangle\|^2$ — while the opposite-sign ancilla-flip effect is small ($\sim 1\text{--}2\%$), so the average retry count stays ~ 6 (three heralds) and ~ 14 (all four) in the presence of noise. The cost is *wall-clock attempts, not coherence depth*: each coherent circuit is a single $\sim 6\mu\text{s}$ pulse, and failed attempts — with all their decoherence — are discarded. Decoherence thus costs accepted-shot fidelity ($\sim 3\%$ below), not yield. Fast, high-fidelity re-preparation of the GKP codeword (the GCR-based scheme of the main text) makes the retries cheap, and the idle state can be kept alive between attempts by stabilizing with a *separate* ancilla in the quadrature orthogonal to the read-out displacement ϵ , leaving the measured quantity untouched.

To quantify the noise budget of one accepted shot, we simulate a single accepted RUS shot on the full GKP codeword ($\langle n \rangle = 6$) with a gate-level Kraus model under the experimental error rates

of Sivak *et al.* (Nature **616**, 50 (2023)): transmon and ancilla $T_1 = T_\phi = 200 \mu\text{s}$, cavity $T_1 = 1000 \mu\text{s}$. Decoherence is applied after each conditional displacement, whose duration equals its amplitude (an amplitude-1 CD takes $1 \mu\text{s}$; the eight gates total $\sim 6 \mu\text{s}$, with the two large position kicks dominating and the small-amplitude corrections nearly instantaneous), and the herald is post-selected. The exact ideal correction sets a noiseless floor of 6×10^{-4} ; decoherence adds

error channel	readout-infidelity increment
transmon decay	+2.0%
transmon dephasing	+0.7%
cavity photon loss	+0.5%
ancilla decay / dephasing	< 0.05% (herald-protected)
all channels	$\approx 3\%$

The accepted-shot readout infidelity is $\approx 3\%$, dominated by transmon *decay* over the $\sim 6 \mu\text{s}$ circuit (the 2π position kick, $|\beta| \approx 2.5$, is by far the longest gate); dephasing and cavity loss are sub-percent because the corrections are fast small-amplitude gates, and *ancilla errors are caught by the herald* — they flip the flag and lower the yield rather than corrupting accepted shots, a built-in error-detection benefit of the flagged construction (and the reason the ancilla channels barely register above). Averaging over the many shots a calibration already takes reduces this further.

Amplitude amplification, by contrast, is shot-efficient but coherence-infeasible. One can instead render a *single* coherent circuit deterministic by amplitude amplification. Let V be the block-encoding of the full correction (the per-pulse gadgets of Sec. 4.2 interleaved with the position kicks), acting on $|\psi\rangle$ and a herald ancilla $|0\rangle_a$:

$$V |0\rangle_a |\psi\rangle = a |0\rangle_a \frac{M|\psi\rangle}{\|M|\psi\rangle\|} + \sqrt{1-a^2} |\Phi^\perp\rangle, \quad a = \frac{\|M|\psi\rangle\|}{\lambda} = \sqrt{P_{\text{succ}}}. \quad (6)$$

Because $|\psi\rangle$ is known, both reflections are available,

$$R_a = I - 2 |0\rangle_a \langle 0|_a \quad (\text{reflect about herald success}), \quad (7)$$

$$R_\psi = V (I - 2 |0\rangle_a |\psi\rangle \langle \psi| \langle 0|_a) V^\dagger \quad (\text{reflect about the input}), \quad (8)$$

and $k \approx \pi/(4 \arcsin a) - \frac{1}{2}$ rounds of the Grover operator $G = -R_\psi R_a$ rotate the amplitude to unity: from $a = 0.27$ (four corrections) and $a = 0.42$ (three) this is $k \approx 3$ and $k \approx 2$ rounds. This is *shot-optimal* — three rounds rather than fourteen retries. But each round requires the input reflection R_ψ , which *un-prepares and re-prepares* $|\psi\rangle$ *coherently*; three rounds therefore stack $\sim 3 \times$ a full GKP preparation into one coherent circuit of $\sim 130 \mu\text{s}$ — comparable to T_1 . A gate-count Kraus budget then gives per-channel exponents $\gamma T \approx \kappa \langle n \rangle T \approx 0.7\text{--}0.9$ and a *total readout infidelity* ≈ 0.98 : the readout is destroyed by decoherence. The determinism is bought with depth — exactly the wrong trade when the error accrues per unit time:

route	coherent depth	attempts	total infidelity
repeat-until-success	$\sim 6 \mu\text{s}$	$\sim 6\text{--}14$	$\approx 3\%$
amplitude amplification	$\sim 130 \mu\text{s}$	1	≈ 0.98

Fixed-point amplitude amplification (Yoder, Low, and Chuang, PRL **113**, 210501 (2014)) removes the need to know a in advance but not the reflections, so it does not change this verdict.

Three single-unitary routes avoid the input reflection R_ψ altogether, each compiling M into one deterministic unitary. The first is a linear combination of unitaries: $e^{c\hat{p}}$ is a Gaussian-weighted

combination of real displacements (Hubbard–Stratonovich), so M is an LCU with 1-norm $\lambda \approx 3.8$, and its PREPARE–SELECT circuit, amplified as above, gives $O(\lambda)$ deterministic queries. The second is multi-copy imaginary-time evolution (Schwartzman *et al.*, arXiv:2603.11208), in which a few copies of $|\psi\rangle$, real-time evolution under H_{eff} , and controlled-SWAPs realize $e^{-\tau H_{\text{eff}}}$ by a *unitary* circuit with no post-selection and no input reflection; the copy count is set by the order needed to reach $\tau\|H_{\text{eff}}\|_{\text{eff}} \sim c/\Delta \sim 1.2$, i.e. a handful. The third is variational QITE (McArdle *et al.*, npj Quantum Inf. **5**, 75 (2019)): since $H_{\text{eff}} = \hat{p}\sigma_\phi$ is low-weight, the ansatz is small and compiles to a fixed deterministic unitary at moderate measurement cost, and Sec. 6 quantifies what such a compilation achieves when used directly as a *stand-alone deterministic readout*. If any of these unitaries has a depth closer to the RUS pulse than to the amplification circuit, it could combine determinism with a tolerable noise budget — the only regime in which a strictly deterministic calibration would be worthwhile.

In scope, every route here needs the input known, re-preparable, or copyable — available in calibration but forbidden for single-shot readout of unknown data by no-cloning. Under realistic decoherence the practical calibration tool is repeat-until-success ($\approx 3\%$ per accepted shot, reduced further by averaging); amplitude amplification, though shot-optimal, is ruled out by its $\sim 130\ \mu\text{s}$ coherent depth.

6 A deterministic readout by variational compilation, and the noise-limited verdict

The calibration routes of Sec. 5 realize the *exact* M . But a fault-tolerant readout does not need the ideal floor — it needs only to sit below the code’s threshold ($\sim 10^{-2}$). This sharpens the question: what is the best *deterministic* readout (a single unitary, no post-selection, no discard, no re-preparation) that a *short* conditional-displacement circuit can achieve? If a shallow unitary already clears threshold, the entire heralding apparatus is unnecessary *for the readout*.

To answer this, we keep the BB1(GCR) position kicks fixed and replace each non-unitary correction by a *deterministic* momentum correction $C_k = e^{-ic_k\hat{p}\sigma_{\nu_k}}$, then optimize the amplitudes and axes $\{c_k, \nu_k\}$ — a QITE-style compilation of the imaginary-time step into a fixed unitary — to minimize the readout error, correctly reading logical-0 (at $\epsilon = 0$) as $+1$ and logical-1 (at $\epsilon = 2u = \sqrt{\pi/2}$) as -1 . We seed from the analytic GCR values — amplitude $c_k = \theta_k\Delta^2/2|\alpha|$ and realizability axis $\sigma_\gamma = \hat{n} \times \hat{\phi}$ (i.e. $\nu_k = \phi_k + \pi/2$) — and then *grow* additional deterministic correction layers to test whether depth helps.

Contrasting the analytic GCR values (what the correction *should* be) with the compiled optimum (what actually minimizes the readout error) exposes two systematic moves (Table 3). The first concerns the axes: the three corrections rotate by $\approx \pi/2$ *off* the GCR-prescribed $\hat{n} \times \hat{\phi}$ axis, landing on the position-kick axis σ_{ϕ_k} itself — which is exactly the axis of the *ideal* non-unitary $M = e^{c\hat{p}\sigma_\phi}$. The best unitary correction thus points along the ideal M ’s own axis, *not* along the realizability-rule axis that makes the physical unitary GCR wash out (Tables 1); re-pointing onto σ_ϕ is precisely how the compiled readout escapes that washout. The target C_4 instead stays near its seed σ_y . The second concerns the amplitudes: the strong 2π correction is *damped* from 4 to 2.9 (a full-strength unitary over-rotates at that angle — point (ii) above), while the target is *boosted* from 1 to 2.5; the two nominally identical π corrections even emerge asymmetric (2.5 vs 1.6). The optimizer keeps the GCR geometry only loosely — it re-points onto the ideal axis and re-weights — but a norm-preserving gate can neither reproduce M ’s contraction nor beat the 1.5×10^{-3} plateau.

The result is a plateau, well above the ideal.

correction	amplitude $c_k [\alpha \Delta^2]$		axis $\nu_k [\pi]$	
	GCR	compiled	GCR	compiled
$C_1 (\pi)$	2	2.5	0.04	0.60
$C_2 (2\pi)$	4	2.9	0.12	0.61
$C_3 (\pi)$	2	1.6	0.04	0.55
$C_4 (\text{target } \pi/2)$	1	2.5	0.50	0.40

Table 3: Deterministic corrections $C_k = e^{-ic_k \hat{p} \sigma_{\nu_k}}$: the analytic GCR prescription versus the QITE-compiled optimum. Amplitudes c_k in units of $|\alpha|\Delta^2$ (the GCR value is $\theta_k \Delta^2 / 2|\alpha| = (2\theta_k/\pi)|\alpha|\Delta^2$, i.e. $\{2, 4, 2, 1\}$); axes ν_k in units of π as headless equatorial directions $\sigma_{\nu} \pmod{\pi}$, with the GCR value $\nu_k = \phi_k + \pi/2 = \hat{n} \times \hat{\phi}$. The corrections rotate $\approx \pi/2$ onto the kick axis σ_{ϕ_k} (the ideal- M axis) and the 2π amplitude is damped $4 \rightarrow 2.9$.

deterministic readout	readout error
bare single conditional displacement	4.5×10^{-2}
bare BB1 (composite, no GCR correction)	7.5×10^{-3}
physical det. BB1(GCR) (σ_y on all four)	0.49 (washes out)
single GCR (target pulse only)	2.6×10^{-3}
optimized deterministic ansatz	1.5×10^{-3}
+1 correction layer	1.5×10^{-3}
+2 correction layers	1.5×10^{-3}
heralded / ideal (post-selected)	5.7×10^{-4}
Helstrom bound	1.1×10^{-6}

The optimized deterministic readout reaches 1.5×10^{-3} — a factor ~ 5 better than bare BB1 (7.5×10^{-3} , the composite pulse the correction is added to) and ~ 1.7 better than a single GCR — and it *escapes* the physical washout (0.49) that a fixed σ_y correction on all four pulses suffers. But it *plateaus*: adding correction layers does not lower it. (This accuracy sweep uses $N_{\text{cav}} = 40$ for optimizer tractability; at the $N_{\text{cav}} = 100$ of the noise simulations single GCR and the optimized ansatz both converge to $\approx 1.4 \times 10^{-3}$, and the plateau persists.) The shallow conditional-displacement class therefore saturates a factor ~ 3 above the herald’s ideal floor and four orders of magnitude above the Helstrom bound. The gap below the finite-energy floor is exactly what the *non-unitary* contraction of Sec. 4 buys and a norm-preserving unitary cannot; a Helstrom-saturating deterministic measurement exists in principle but requires a deep, non-local circuit outside this local ansatz. This is the note’s central conclusion reached from the deterministic side: no short unitary matches the heralded floor.

These coherent floors are, however, largely academic under realistic decoherence, where accounting for gate *durations* overturns the ranking altogether — length, not accuracy, decides. A conditional displacement of amplitude $|\beta|$ takes $|\beta| \mu\text{s}$ (an amplitude-1 CD is $1 \mu\text{s}$), so a readout’s cost is set by its *position kicks*; the corrections are tiny-amplitude ($|\beta| \lesssim 0.3$) and nearly free. The full BB1(GCR) is dominated by its 2π kick ($|\beta| \approx 2.5$, i.e. $2.5 \mu\text{s}$) — a gate it *shares with bare BB1* — whereas single GCR omits it, using only the $\theta_t = \pi/2$ kick. At the rates of Sec. 5:

readout	circuit time	coherent floor	noise	total
single GCR	$0.7\ \mu\text{s}$	1.4×10^{-3}	+0.2%	$\approx 0.4\%$
bare BB1	$5.6\ \mu\text{s}$	6.4×10^{-3}	+2.6%	$\approx 3.2\%$
BB1(GCR), deterministic	$6.3\ \mu\text{s}$	1.5×10^{-3}	+3.2%	$\approx 3.2\%$
BB1(GCR), heralded	$6.3\ \mu\text{s}$	5.7×10^{-4}	+3.2%	$\approx 3.2\%$ (17% yield)

Two things follow. First, BB1(GCR) is essentially *bare BB1* under noise: the GCR correction lowers the coherent floor by an order of magnitude ($6.4 \times 10^{-3} \rightarrow 6 \times 10^{-4}$) at almost no extra circuit time (+0.7 μs , +12%) — the corrections are essentially free — but the $\sim 3\%$ noise, set by the *shared* kicks, masks the gain entirely, so both land at $\approx 3.2\%$. Second, single GCR wins by $\sim 8\times$: dropping the long 2π kick makes it a 0.7 μs circuit, so decoherence adds only $\sim 0.2\%$, and its coherent floor (1.4×10^{-3}) already *equals* the deterministic plateau, so it sacrifices no accuracy to get there. Under realistic noise the shortest adequate readout dominates.

The practical choice for a *noisy* GKP readout is therefore **single GCR** — $\approx 0.4\%$, deterministic, discard-free, and the shortest circuit that clears threshold. The elaborate BB1(GCR) machinery (the 2π correction, heralding, post-selection) delivers its lower coherent floor only in a low-noise limit today’s hardware does not reach: its advantage over bare BB1 and single GCR stays masked until the per-shot decoherence drops below the coherent gap, $\sim 10^{-3}$ (an order of magnitude better coherence than assumed here). BB1(GCR)’s roles are thus (a) the high-coherence / noiseless-limit readout; (b) the certification tool that fixes the deterministic readout’s floor; and (c) the high-fidelity primitive for known-state preparation and calibration (Sec. 5), where re-preparation is free and the 5.7×10^{-4} – 4×10^{-7} accuracy is the whole point. For end-of-line data readout on current devices, single GCR is the better choice.

7 Readout under realistic noise: master-equation simulation

The gate-count budget of Sec. 6 is confirmed and extended by a full master-equation (**mesolve**) simulation at $N_{\text{cav}} = 400$. Each conditional displacement of amplitude $|\beta|$ is evolved for a duration $|\beta|\ \mu\text{s}$ (amplitude-1 CD = 1 μs) under the experimental collapse operators (transmon and ancilla decay and dephasing $T_1 = T_\phi = 200\ \mu\text{s}$; oscillator photon loss $T_1 = 1000\ \mu\text{s}$). We simulate seven readouts — single GCR, the QITE-compiled deterministic BB1(GCR), bare BB1, GCR-BB1 (bare BB1 preceded by a single QITE-optimized momentum pre-correction), two *block* variants that prepend four momentum pre-corrections (block A, block B), and the heralded BB1(GCR) — isolating each channel and the total across the two Fig-3 displacement-error windows (Fig. 6; readout error in Fig. 4, oscillator infidelity in Fig. 5).

Consider the readout error first. At the operating point, under all channels,

readout	full-noise error	dominant channel
single GCR	3.3×10^{-3}	transmon decay
BB1(GCR), deterministic	2.2×10^{-2}	transmon decay
GCR-BB1 (one pre-correction)	2.4×10^{-2}	transmon decay
GCR-BB1 (block A)	2.4×10^{-2}	transmon decay
GCR-BB1 (block B)	2.4×10^{-2}	transmon decay
bare BB1	2.8×10^{-2}	transmon decay
heralded BB1(GCR)	3.7×10^{-2}	transmon decay

confirming the verdict of Sec. 6: single GCR is $\sim 8\times$ more accurate than the full-length schemes because it is the shortest circuit (it omits the long 2π kick), and the bit error is set by transmon

decoherence over the circuit — oscillator loss barely touches the readout qubit ($+6 \times 10^{-4}$ for GCR).

The oscillator back-action turns out to be a *recoverable* displacement rather than genuine damage. The post-read oscillator stays nearly *pure* (impurity $1 - \text{Tr}\rho^2 \approx 1\text{--}18\%$, growing with circuit length), yet its *raw* overlap with the input codeword is only $\sim 0.5\%$ (fidelity $F \approx 0.07$). These are reconciled by a single fact: the read applies a large, *state-agnostic* displacement — the conditional-displacement pulses kick the oscillator by an amount fixed by the qubit, *independent of the oscillator’s location*. Calibrating this displacement *once* on the no-error logical-0 readout output,

$$\beta_{\text{corr}} = +i \frac{1}{2} \sqrt{\pi/2} \quad (\text{a pure momentum shift}), \quad (9)$$

and applying $D(\beta_{\text{corr}})$ to *every* input — the corrective displacement used in the readout itself — recovers the oscillator fidelity from $F \approx 0.07$ to

readout	noiseless F	full-noise F
single GCR	0.974	0.971
GCR-BB1 (one pre-correction)	0.977	0.953
GCR-BB1 (block A)	0.977	0.951
GCR-BB1 (block B)	0.976	0.950
bare BB1	0.975	0.951
BB1(GCR), deterministic	0.960	0.933
heralded BB1(GCR)	0.974	0.926

flat across the ϵ window. So the oscillator is $\sim 93\text{--}97\%$ preserved per read after one calibrated corrective displacement — the apparent near-orthogonality was entirely this recoverable kick, not decoherence — and, as with the readout error, *single GCR preserves it best under noise* (0.971), its short circuit accruing the least decoherence while the longer BB1(GCR) drops to 0.933. The residual $\sim 5\%$ (GCR) is a genuinely non-Gaussian distortion: neither a further displacement nor a squeeze removes it (the optimal corrective squeeze is exactly zero), consistent with the irreducible measurement back-action of extracting the bit. Appending small-Big-small stabilization rounds to the post-read state does *not* reduce this residual — it is not a lattice displacement, the only error such stabilization corrects — so the corrective displacement is the operative and sufficient mechanism. The only further gain available is combining *repeated* GCR reads (two reads log-odds-combined reach $\sim 2 \times 10^{-6}$).

A fourth deterministic option sharpens the interpretation of what the GCR correction actually buys. GCR-BB1 keeps the four bare BB1 kicks but prepends a *single* conditional momentum displacement $e^{-ic\hat{p}\sigma_y}$, whose two parameters we optimize by the same QITE search used for the interleaved corrections (Sec. 6). The optimizer improves this lone pre-correction sharply — readout error $1.6 \times 10^{-2} \rightarrow 2.3 \times 10^{-3}$, a $\sim 7\times$ gain — by slightly enlarging its amplitude and tilting its axis $\sim 9^\circ$ off σ_y , the same rotation-toward-the-kick-axis we saw in Table 3. Because that pre-correction is a $\sim 0.02 \mu\text{s}$ gate against the $\sim 5.6 \mu\text{s}$ of kicks it *shares with bare BB1*, GCR-BB1 carries essentially bare BB1’s length, and hence its decoherence floor — yet it sits strictly above bare BB1 everywhere: lower readout error (noiseless 2.1×10^{-3} vs 6.1×10^{-3} ; full-noise 2.4×10^{-2} vs 2.8×10^{-2}) and higher fidelity in both windows. Most tellingly, in the discriminating logical-1 window it *beats the fully interleaved deterministic BB1(GCR)* ($F = 0.915$ vs 0.903 under noise), because it pays for one correction rather than four: a single well-placed QITE-tuned pre-kick captures most of the coherent benefit at a fraction of the added machinery. Under noise it remains, like every full-length scheme, behind single GCR — but it is the clearest statement that the corrections want to be *few and well-placed*, not numerous.

This invites the converse question: whether a *block* of well-placed pre-corrections, rather than a single one, can recover the interleaved scheme’s coherent floor without paying its interleaved length. We test two such blocks that prepend *four* momentum corrections ahead of the bare BB1 kicks — block A, a free QITE-optimized momentum-BB1 block, and block B, the interleaved BB1(GCR) corrections analytically conjugated to the front and then QITE-polished. Both slash the noiseless readout error to 6.7×10^{-5} (A) and 4.5×10^{-5} (B), recovering $\sim 30\text{--}47\times$ of the single-pre-correction gap yet remaining $\sim 5\text{--}7\times$ above the fully interleaved BB1(GCR) (9.8×10^{-6}). Under realistic noise, however, the blocks carry the same full BB1 kick length and hence the same $\sim 2.4 \times 10^{-2}$ decoherence floor as GCR-BB1 and bare BB1 — the coherent gain washes out entirely. The block therefore trades a slightly longer prefix for a markedly lower *noiseless* floor without matching interleaving, and confirms once more that under decoherence the corrections that matter are few and well-placed, not numerous.

One might expect heralding to help under noise through its built-in post-selection, but it does not, and the reason is structural. The herald measures the *ancilla*, whereas the readout bit lives on the *readout qubit*. Under noise the bit error is dominated by transmon decoherence during the long position kicks, where the ancilla is idle and uncoupled — so these errors are *uncorrelated* with the herald and cannot be post-selected away. Post-selection rejects only the sub-dominant correction-gadget errors: it prevents the heralded readout from being *worse*, but cannot make it *better* than the deterministic read. Faithful post-selected simulations confirm the accepted-shot error stays $\gtrsim$ the deterministic value at a large ($\gtrsim 75\%$) yield cost. Heralding buys error *detection*, not readout accuracy.

8 Reading a pre-damaged codeword: when readout engineering stops helping

The noise analysis of Sec. 7 charged decoherence only to the readout *circuit*: the codeword entering each read is pristine, and the error is what the finite-duration pulses accrue. A stored logical qubit faces the complementary hazard — it idles in the oscillator, accumulating photon loss and dephasing, *before* the readout is ever triggered. We now ask how each protocol reads a codeword that has *already* been damaged, isolating this input-damage axis from the readout-circuit noise studied above.

Noise model. The experiment has two temporally separate stages, and it is important to keep their noise sources distinct. We prepare the canonical no-error logical-0 and logical-1 codewords — the $\epsilon = 0$ finite-energy GKP states of Sec. 7, built with envelope $\Delta = 0.34$ in an $N_{\text{cav}} = 400$ Fock space — and first subject the *bare oscillator* to an idle decoherence stage, then read the damaged state out. During the idle stage the transmon is uncoupled and the system Hamiltonian is set to zero, so the density matrix evolves under a purely dissipative Lindblad master equation with a single collapse operator $\hat{L}$,

$$\dot{\rho} = \mathcal{D}[\hat{L}]\rho \equiv \hat{L}\rho\hat{L}^\dagger - \frac{1}{2}\{\hat{L}^\dagger\hat{L}, \rho\}, \quad (10)$$

integrated for an idle time $\tau \in [0, 1000] \mu\text{s}$. Because Eq. (10) is time-independent we integrate it in fixed $20 \mu\text{s}$ substeps, which bounds the solver’s local error without changing the channel. We compare two channels on a common τ -axis, each taken as the *only* dissipator so that its effect is seen in isolation:

- *oscillator photon loss*, $\hat{L} = \sqrt{\kappa} \hat{a}$ with $\kappa = 1/T_1^{\text{cav}} = 1/1000 \mu\text{s}^{-1}$ — the cavity- T_1 rate of the Sivak *et al.* model used throughout this note. Amplitude damping pulls amplitude toward

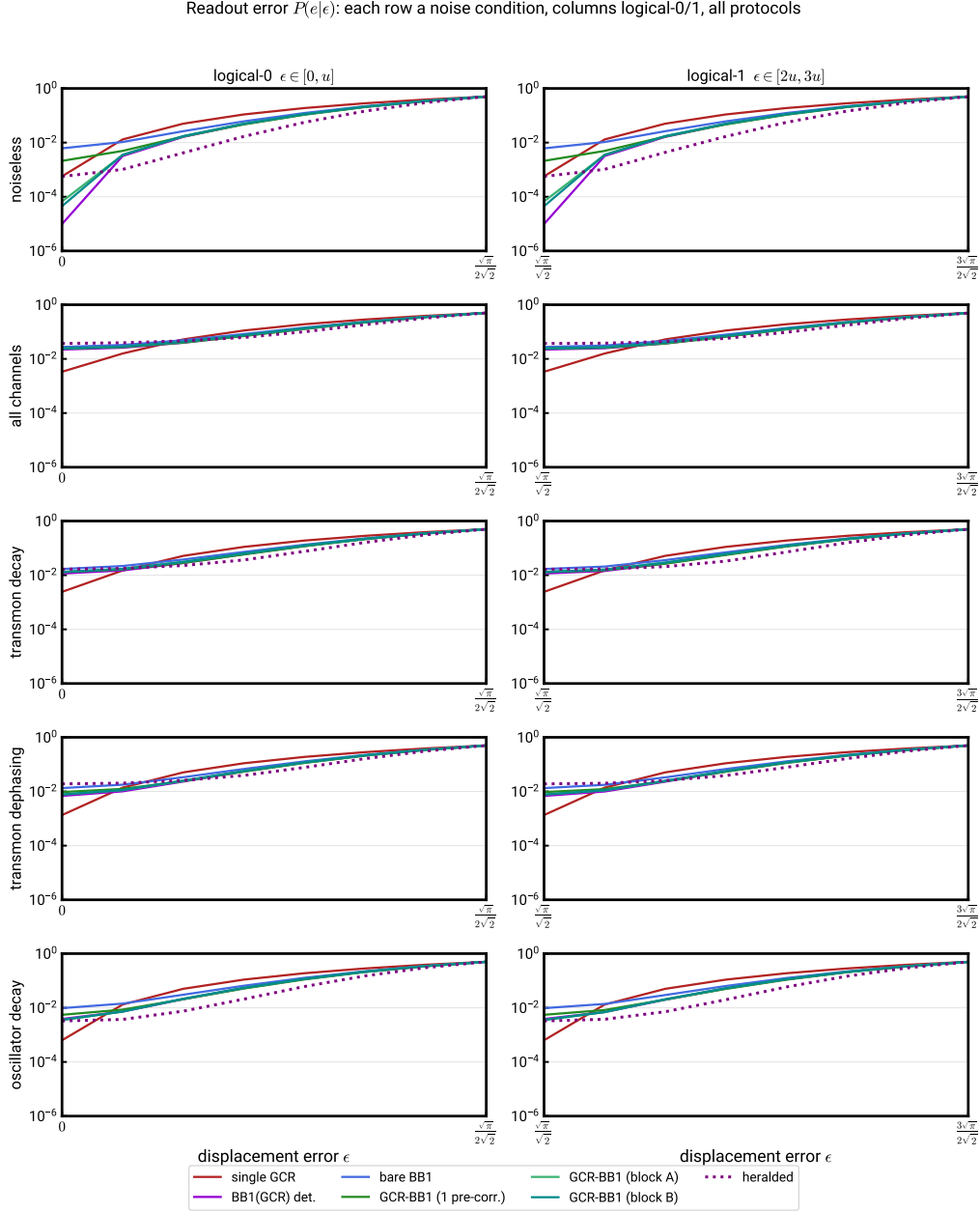


Figure 4: Master-equation readout error $P(e|\epsilon)$ vs. displacement error ϵ for the seven readouts: single GCR (red), deterministic BB1(GCR) (violet), bare BB1 (blue), GCR-BB1 (teal), GCR-BB1 block A (mediumseagreen) and block B (darkcyan), heralded (purple, dotted). Rows are noise settings (noiseless; all channels; transmon decay; transmon dephasing; oscillator decay); columns are the logical-0 ($\epsilon \in [0, u]$) and logical-1 ($\epsilon \in [2u, 3u]$) windows, $u = \sqrt{\pi}/(2\sqrt{2})$. Noiseless, BB1(GCR) reaches $\sim 10^{-5}$ (near-Helstrom); under decoherence all full-length schemes converge to a $\sim 2-3 \times 10^{-2}$ floor set by circuit time, while the short single-GCR read stays $\sim 8\times$ lower. GCR-BB1 (teal) and its block variants (green/cyan) track bare BB1 but lie consistently below it; the blocks reach the lowest noiseless floor of the deterministic schemes.

Oscillator infidelity $1 - F$ (fixed corrective displacement): each row a noise condition, columns logical-0/1, all protocols

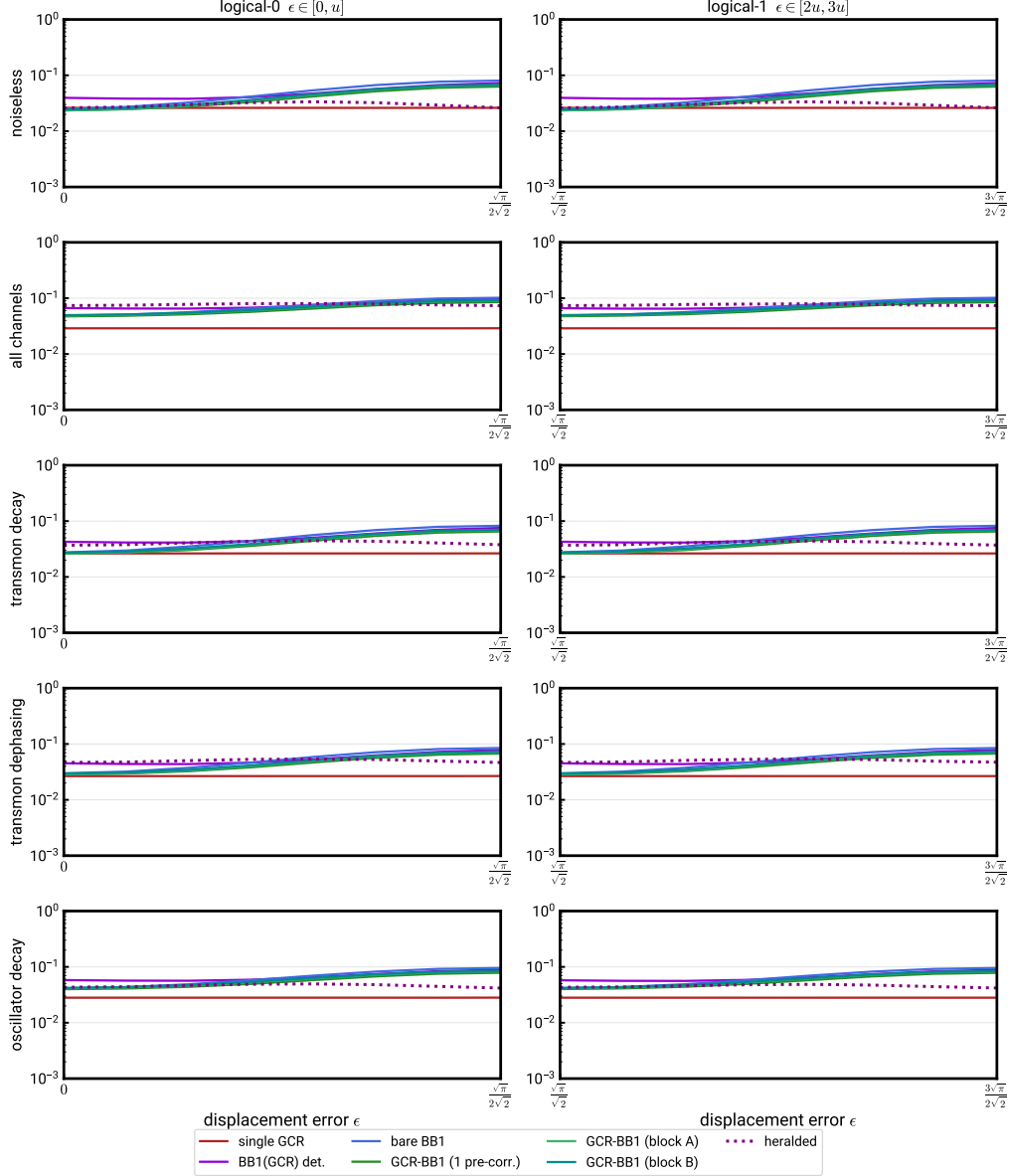


Figure 5: Post-read oscillator infidelity $1 - F$ to the input codeword, *after* the single calibrated corrective displacement $D(\beta_{\text{corr}})$, for the same seven readouts and layout as Fig. 4. The residual is nearly flat in ϵ and ordered by circuit length: single GCR ($\approx 2.9 \times 10^{-2}$) < bare BB1 ($\approx 4.9 \times 10^{-2}$) < deterministic BB1(GCR) ($\approx 6.7 \times 10^{-2}$) < heralded ($\approx 7.4 \times 10^{-2}$). GCR-BB1 (teal) sits just below bare BB1 in the logical-1 window, with the block variants alongside it. The floor is the irreducible non-Gaussian back-action of extracting the bit, not a lattice displacement.

the vacuum, so it *contracts and blurs* the quadrature comb that carries the logical bit;

- *oscillator dephasing*, $\hat{L} = \sqrt{\kappa_\phi} \hat{a}^\dagger \hat{a}$ with $\kappa_\phi = 1/5000 \mu s^{-1}$. The photon-number operator $\hat{a}^\dagger \hat{a}$ commutes with energy, so this channel *smears the comb in phase* while leaving the lattice spacing — and hence the mean photon number — intact.

The Sivak model specifies no intrinsic cavity dephasing, so κ_ϕ has no hardware calibration; we deliberately set it a factor of five slower than photon loss and treat the dephasing column as a *qualitative* comparison of channel character, not a device prediction. After the idle stage the damaged oscillator is read out by each protocol with *noiseless* pulses — the readout circuit itself accrues no decoherence — so that every residual error is attributable to the pre-existing input damage alone. This cleanly separates the input-damage axis studied here from the readout-circuit-noise axis of Sec. 7, whose complementary limit was a pristine input read by noisy pulses. Figure 7 tracks the readout error $P(e)$ and the post-read oscillator infidelity $1 - F$ (after the same single calibrated corrective displacement of Sec. 7) against τ , averaged over the two codewords, which are symmetric under both channels.

The dominant message is a *collapse of the protocol hierarchy*. At $\tau = 0$ the seven curves reproduce their noiseless operating points, spanning more than two orders of magnitude — from BB1(GCR) and the two blocks at $\sim 10^{-5}$ – 10^{-4} , up through single GCR (6×10^{-4}), GCR-BB1 (2.1×10^{-3}), to bare BB1 (6.1×10^{-3}). Idle photon loss erases this separation almost immediately: the spread is already inside a factor of ~ 2 by $\tau \approx 170 \mu s$ (a sixth of the oscillator T_1), inside 10% by $\tau \approx 500 \mu s$, and by $\tau = 1000 \mu s$ all seven agree to within 1%, clustered at $P(e) \approx 0.46$ — just short of the $P(e) = \frac{1}{2}$ random-guess limit. The reason is physical: the logical bit is carried by the position-quadrature *comb*, and photon loss contracts and blurs that comb before any pulse acts. Once the lattice information has leaked out of the input, no readout can recover it, and the choice of correction gadget becomes irrelevant. Readout engineering buys accuracy only in the regime where the stored state is still intact.

Dephasing is far gentler at these rates: it smears the comb in phase without contracting the lattice, and at $\kappa_\phi = 1/5000 \mu s^{-1}$ the error climbs only to $P(e) \approx 0.22$ by $\tau = 1000 \mu s$, with the hierarchy again collapsing — the seven protocols lie within 5% of one another past $\tau \approx 330 \mu s$.

A subtler trend inverts the verdict of Sec. 7. There, single GCR was the most accurate readout because its short circuit accrued the least decoherence. Here, against a *pre-damaged* input, single GCR is instead the *least* tolerant at intermediate damage: at $\tau \approx 170 \mu s$ it reads at $P(e) = 0.087$ (photon loss) and 0.076 (dephasing), while every BB1-structured scheme sits at 0.040–0.068 and 0.061–0.074 respectively. A single correction gives the narrowest correctable region, so a partially-decayed input spills past it fastest; the multi-correction schemes, whose entire purpose is to widen that region, absorb more of the damage-induced spread and degrade more gracefully. The effect is modest and vanishes once the damage is severe, but it is the mirror image of the readout-noise story: single GCR wins when the bottleneck is the readout *circuit*, while the correction-heavy schemes edge ahead when the bottleneck is a *damaged input*.

The post-read oscillator infidelity tells the same story from the state’s side, only more starkly. At $\tau = 0$ it reproduces the readout-only floors of Sec. 7’s fidelity table — $1 - F \approx 0.023$ for GCR-BB1 and the blocks, rising to 0.040 for the interleaved BB1(GCR), the same modest ordering set by the single calibrated corrective displacement. Under idle damage it climbs monotonically: photon loss drives it to $1 - F \approx 0.65$ by $\tau = 1000 \mu s$, dephasing only to ≈ 0.44 , again the decay-dominates-dephasing hierarchy. But the protocol spread here is even tighter than for the readout bit — the seven schemes stay within $\lesssim 0.016$ of one another at *every* damage time, so the surviving oscillator overlap is set almost entirely by the input damage and barely at all by which readout was applied. The one qualitative wrinkle inverts the $P(e)$ ranking once more: the heralded protocol, the

most accurate on the readout bit, leaves the *least* faithful oscillator state (by ~ 0.01), because its mid-circuit projections perturb the surviving comb slightly more than the purely-unitary schemes — heralding optimizes the logical readout, not the state it leaves behind.

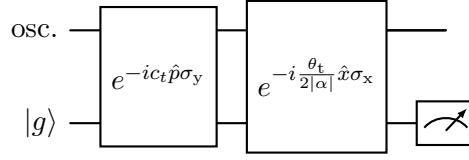
Both hazards at once. The noiseless-readout panels isolate input damage, but a real device suffers both hazards together: the stored codeword decays *and* the pulses that read it are themselves noisy. Figure 8 repeats the damage sweep with the readout circuit now running under each of the five noise conditions of Sec. 7 (all channels, transmon decay, transmon dephasing, oscillator decay, oscillator dephasing), so that the two axes act simultaneously. The result sharpens rather than softens the verdict.

Readout-circuit noise reshuffles the $\tau = 0$ operating points and *inverts* the noiseless ranking, exactly as Sec. 7 predicts. Under the all-channels condition the interleaved BB1(GCR), which reads at $\sim 10^{-5}$ with noiseless pulses, is lifted to $P(e) = 2.2 \times 10^{-2}$ — now among the *worst* protocols — while single GCR, whose short circuit accrues the least decoherence, sits lowest at 3.3×10^{-3} , and the heralded scheme trails at 3.7×10^{-2} . The correction-heavy circuits pay for their length in readout noise: the very structure that buys a near-Helstrom noiseless floor becomes a liability the moment the pulses are lossy. Weakening the readout channel (transmon dephasing, oscillator dephasing) lowers these floors uniformly but preserves the ordering, with single GCR always the most accurate read at $\tau = 0$.

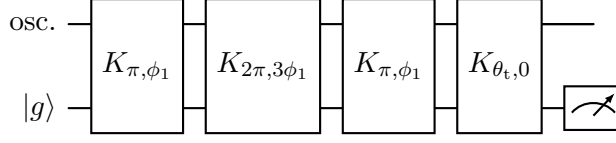
The interplay of the two axes produces a clean *crossover*. Single GCR is the best readout at $\tau = 0$ but the least tolerant of a decayed input; as damage accumulates its curve rises through the pack, and by $\tau \approx 167 \mu\text{s}$ under all-channels readout it is the *worst* of the seven ($P(e) = 0.090$, against 0.075 for the correction-heavy schemes) — the mirror image of its $\tau = 0$ lead. This is the two sections meeting in a single figure: single GCR wins when the bottleneck is the readout *circuit* and loses when it is a *damaged input*, and the changeover sits near $\tau \sim 150 \mu\text{s}$ for these rates. Past that point every protocol collapses together exactly as in the noiseless case.

Most telling is what the readout noise does *not* change. At $\tau = 1000 \mu\text{s}$ all seven protocols cluster at $P(e) = 0.467$ under photon loss and ≈ 0.23 under dephasing — and these tail values are identical, to better than 10^{-3} , across *all six* readout conditions including the noiseless one. Once the input is destroyed, whatever noise the readout pulses add is a negligible perturbation on top of information that has already leaked out of the oscillator. The post-read oscillator infidelity (Fig. 9) tells the same story from the state’s side: the $\tau = 0$ floors are lifted to $1-7 \times 10^{-2}$ depending on the readout condition — the interleaved BB1(GCR) again paying the most, up to 7.1×10^{-2} under oscillator-dephasing readout — yet the tails converge to $1 - F \approx 0.65$ (loss) and ≈ 0.44 (dephasing) essentially independent of which pulses were used. Adding realistic readout noise therefore does not rescue the hierarchy — it only confirms that beyond a modest damage threshold the outcome is a *memory* failure that no readout, noisy or clean, can undo.

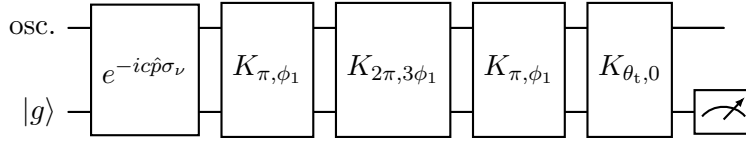
Together with Sec. 7, this bounds the operational value of the whole construction. The elaborate corrections earn their keep against readout-circuit noise, and buy a modest graceful-degradation margin against a lightly-damaged input; but a heavily-damaged codeword is a *memory* failure, not a readout failure, and is beyond the reach of any readout. The remedy there is a better oscillator or more frequent stabilization — not a longer pulse train.



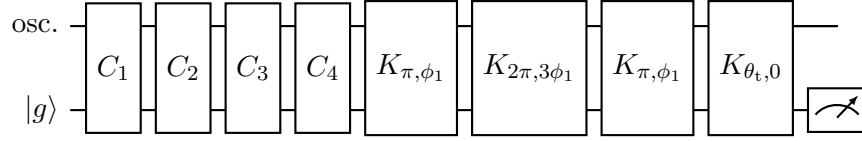
(a) single GCR: one momentum pre-correction + one position kick.



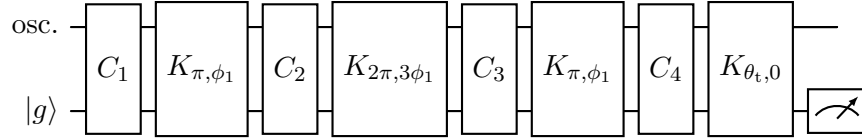
(b) bare BB1: four position kicks $K_{\theta, \phi} = e^{-i \frac{\theta}{2|\alpha|} \hat{x} \sigma_\phi}$, no correction.



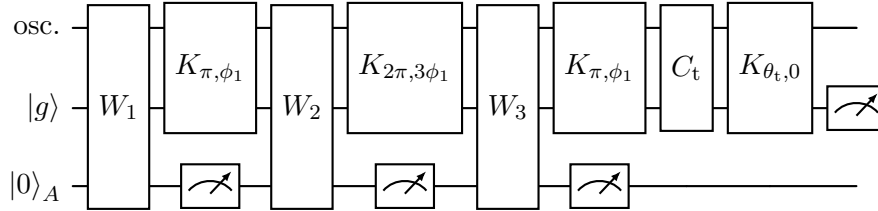
(c) GCR-BB1: a *single* QITE-optimized momentum pre-correction $e^{-ic \hat{p} \sigma_\nu}$ before the four *bare* BB1 kicks.



(d) GCR-BB1 (block): a *block* of four momentum pre-corrections $C_k = e^{-ic_k \hat{p} \sigma_{\nu_k}}$ before the four *bare* BB1 kicks (block A: free QITE optimization; block B: interleaved BB1(GCR) corrections conjugated to the front, then QITE-polished — same circuit structure, different parameters).



(e) deterministic BB1(GCR): each pulse is a momentum pre-correction $C_k = e^{-ic_k \hat{p} \sigma_{\nu_k}}$ (QITE-optimized axis ν_k) followed by its position kick.



(f) heralded BB1(GCR): each of the three strong corrections is a block-encoding gadget $W_k = e^{-ic_k \hat{p} \sigma_{\phi_k} \otimes \sigma_y}$ (Fig. 1) acting on oscillator, system qubit, and a fresh ancilla, post-selected on ancilla outcome 0 (meters); the deterministic target $C_t = e^{-ic_t \hat{p} \sigma_y}$ closes the sequence. A corrective displacement $D(\beta_{\text{corr}})$ (not shown) is appended to all cases to recover the oscillator.

Figure 6: Exact conditional-displacement sequences for the seven simulated readouts. All gates act on oscillator+system-qubit; only the heralded readout uses the ancilla A . Time runs left to right; the readout bit is the final system-qubit measurement.

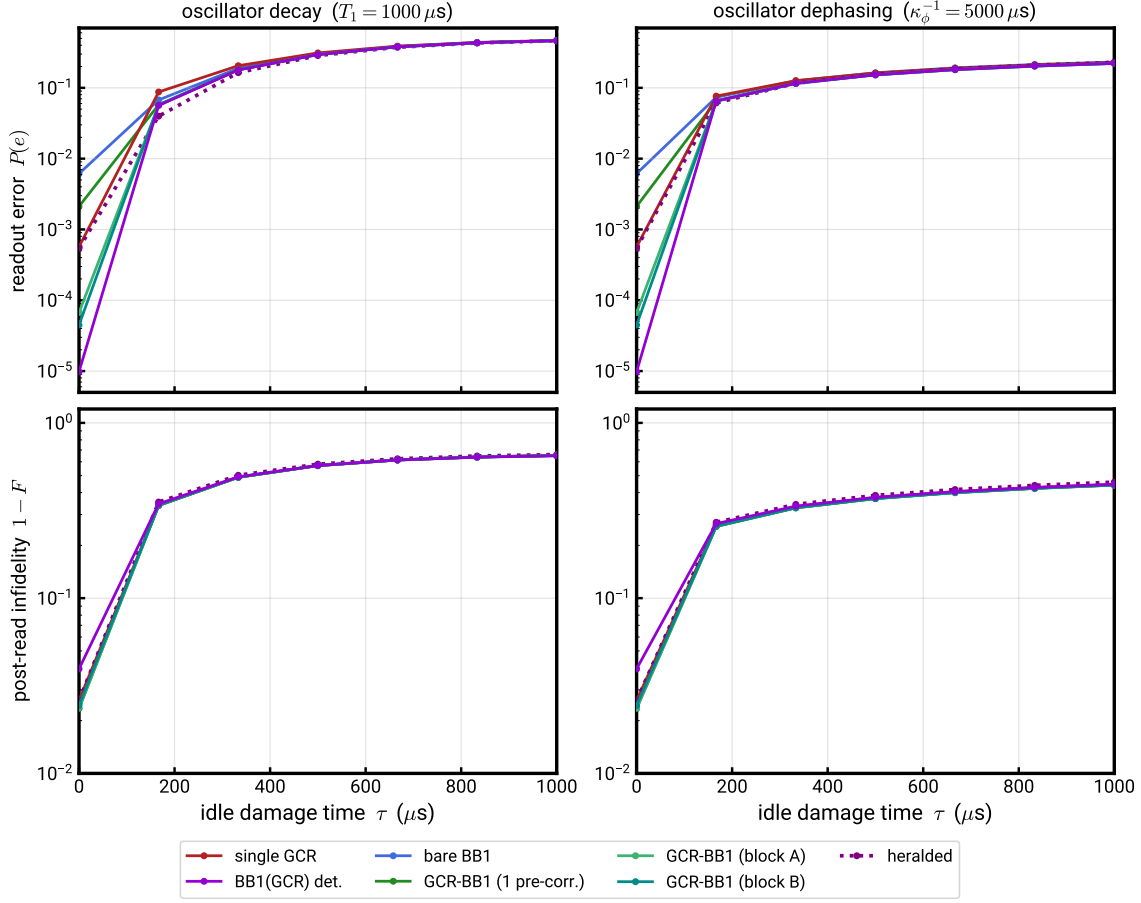


Figure 7: Reading a pre-damaged codeword. Undisplaced no-error logical-0/1 GKP codewords are idled under a pure oscillator channel for time τ and then read with *noiseless* pulses, isolating the effect of input damage from readout-circuit noise. Columns: oscillator photon loss ($T_1 = 1000 \mu\text{s}$, left) and oscillator dephasing ($\kappa_\phi^{-1} = 5000 \mu\text{s}$, right). Top row: readout error $P(e)$; bottom row: post-read oscillator infidelity $1 - F$ after a single calibrated corrective displacement. Each panel overlays the seven readout protocols (colors as in Figs. 4 and 5); curves are averaged over the two codewords. The $\tau = 0$ points reproduce the noiseless operating points of Sec. 7; under photon loss the $> 100\times$ spread among protocols collapses to within 1% by $\tau = 1000 \mu\text{s}$, as the lattice information leaks out of the input faster than any readout can extract it.

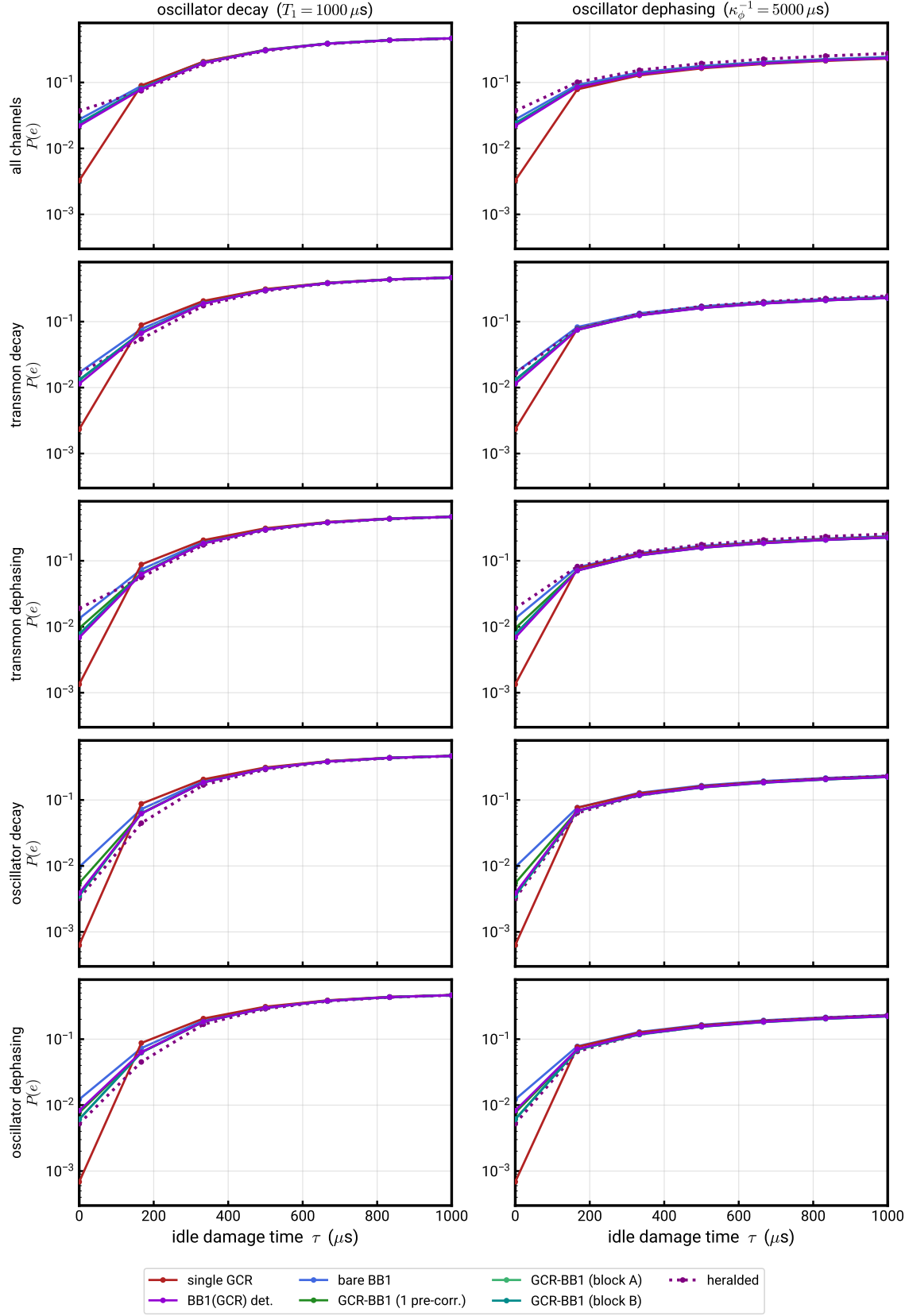


Figure 8: **Reading a pre-damaged codeword with *noisy* pulses.** As Fig. 7, but the readout circuit now accrues decoherence under each of the five noise conditions of Sec. 7 (rows), for the two idle-damage channels (columns: oscillator photon loss, left; oscillator dephasing, right). Curves are the readout error $P(e)$ for the seven protocols, averaged over the two codewords. Readout-circuit noise lifts the $\tau = 0$ floors and *inverts* the protocol ranking there — the noiseless-best interleaved BB1(GCR) is now among the worst, single GCR the best — but the large- τ tail is set almost

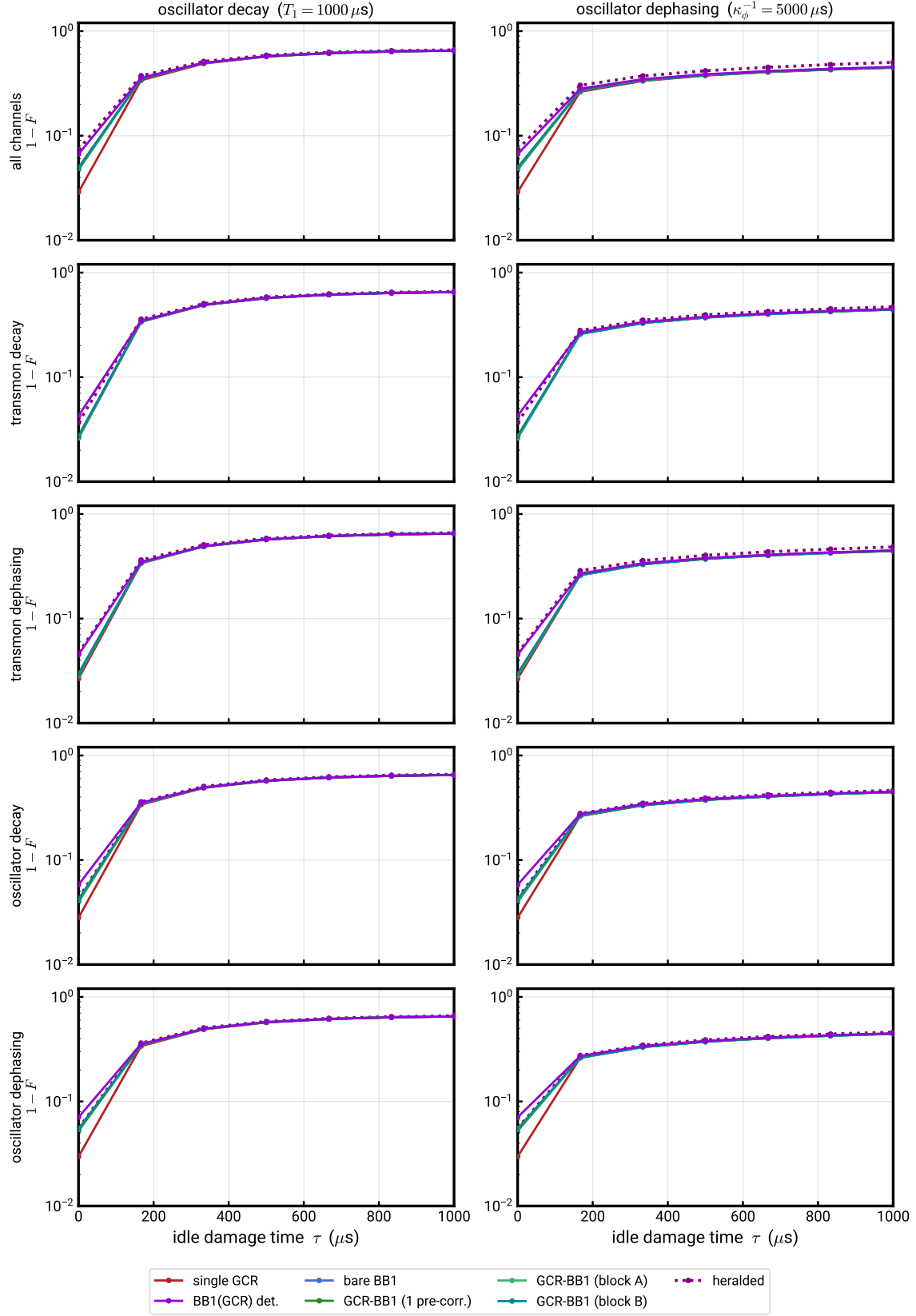


Figure 9: **Post-read oscillator infidelity with noisy pulses.** Companion to Fig. 8 on the identical grid: post-read oscillator infidelity $1 - F$ (after the single calibrated corrective displacement) for the five readout-noise conditions (rows) and the two idle-damage channels (columns), seven protocols overlaid. As for the readout bit, the readout condition sets only the lightly-damaged floors; the heavily-damaged tails ($1 - F \approx 0.65$ under photon loss, ≈ 0.44 under dephasing) are fixed by the input damage and are common to all conditions.

9 Reading an actively stabilized codeword: noisy SBS rounds before readout

Section 8 let the codeword idle passively. A real logical memory does the opposite: it is *actively stabilized*, its lattice periodically re-sharpened by an error-correction protocol that itself runs on the same imperfect hardware. The canonical GKP stabilizer is the small-big-small (SBS) round of Royer *et al.* and Campagne-Ibarcq *et al.*, realized on the Sivak *et al.* device: a sequence of conditional displacements interleaved with ancilla rotations that pumps entropy out of the oscillator and pushes the state back toward the code manifold. But each round is a finite-duration circuit coupling the oscillator to a lossy ancilla, so stabilization is a double-edged tool — it removes lattice error while injecting fresh decoherence. We ask how this trade-off interacts with the readout: after m rounds of *noisy* SBS, how faithfully does each of the seven protocols read the stabilized codeword, and how does that degrade as the rounds accumulate?

Model. We use the authentic SBS protocol of the Sivak *et al.* device — a σ_z -conditioned time evolution under the marker Hamiltonians $\hat{H}_x = \hat{\sigma}_z \otimes a_1 \hat{x}$ and $\hat{H}_p = -\hat{\sigma}_z \otimes a_1 \hat{p}$, with an $R_x(\pi/2)$ rotation conjugating the big step, acting on a finite-energy *binomial* GKP codeword ($\Delta = 0.34$, $N_{\text{cav}} = 200$). The fresh $m = 0$ codeword is the fixed point of noiseless cooling: we run the stabilizer forward from vacuum for twenty rounds to obtain the stabilized logical-0 state $G_{\text{SBS},0}$, and take that as the input to the experiment. A single SBS round of this protocol exchanges the two logical branches — it is a stable two-cycle, $G_{\text{SBS},0} \leftrightarrow G_{\text{SBS},1}$, with the fidelity returning to $\gtrsim 0.98$ after every second round — so we sample an even number of rounds $m \in \{0, 2, 4, 6, 8, 10\}$, for which the encoding always returns to logical 0 while having absorbed the decoherence of m physical stabilization circuits.¹

Noise enters the SBS rounds through the same collapse operators used throughout this note. During the marker evolutions the master equation carries, according to the condition under test, transmon decay $\sqrt{\gamma} \hat{\sigma}_-$ ($\gamma = 1/200 \mu\text{s}^{-1}$), transmon dephasing $\sqrt{\gamma_\phi/2} \hat{\sigma}_z$ ($\gamma_\phi = 1/200 \mu\text{s}^{-1}$), oscillator photon loss $\sqrt{\kappa} \hat{a}$ ($\kappa = 1/1000 \mu\text{s}^{-1}$), or oscillator dephasing $\sqrt{\kappa_\phi} \hat{a}^\dagger \hat{a}$ ($\kappa_\phi = 1/5000 \mu\text{s}^{-1}$, the same qualitative rate as in Sec. 8). We run five separate stabilization experiments, one per SBS-noise condition: *all channels* together (transmon decay + transmon dephasing + oscillator loss), and each of transmon decay, transmon dephasing, oscillator decay, and oscillator dephasing in isolation. In every experiment the stabilized state at each m is then read out under all six readout-circuit conditions of Sec. 7 — noiseless, all channels, transmon decay, transmon dephasing, oscillator decay, oscillator dephasing — so that the SBS-noise axis and the readout-noise axis are cleanly separated. As before we report the readout error $P(e)$ and the post-read oscillator infidelity $1 - F$ after the single calibrated corrective displacement, against the pristine $m = 0$ codeword; the analysis is for logical 0.

A note on the vertical scale. Because a genuine, non-collapsing SBS stabilizer is available only in the device’s native *binomial*-GKP, $i = 2$ quadrature convention, this section uses that codeword rather than the Gaussian-comb encoding of Secs. 7–8. The stabilized codeword is correspondingly softer than the idealized comb — even the freshly-cooled $m = 0$ state reads at $P(e) \approx 0.08$ under noiseless pulses, the validated operating point of this SBS stabilizer, rather than the $\lesssim 10^{-3}$ floors of Sec. 7. The comparison of interest here is therefore *relative*: how each protocol and each noise

¹This parity exchange is a feature of a single SBS round, not a defect: it is why the cooling trajectory alternates between the $\mu = 0$ - and $\mu = 1$ -like states on consecutive iterations. Odd m would simply read the logical-1 branch of the same data.

channel move the operating point as the rounds accumulate, not the absolute floor, which is set by the encoding rather than the readout gadget.

Heralding does not transfer to the stabilized codeword. One protocol must be set aside at the outset. Unlike the six deterministic protocols — whose corrections are conditional *displacements* that act relative to the state and therefore inherit whatever lattice the codeword carries — the heralded scheme relies on a fixed, non-covariant momentum *filter* whose geometry is pinned to the Gaussian-comb lattice of Secs. 7–8. On the binomial-GKP fixed point of this SBS stabilizer, that filter is mis-registered: a scan of the input displacement over a full two-cell window fails to bring the heralded readout below $P(e) \approx 0.47$ anywhere on the real axis, and momentum-axis displacements drive the post-selection norm through zero (the heralding branch is rejected) rather than recovering it. In other words, the heralded gadget would have to be re-derived from scratch for each encoding’s lattice; a rigid recalibration cannot rescue it. We therefore report the *six deterministic* protocols here and note only that the heralded scheme, alone among the seven, is not portable across the encoding change — a concrete cost of building readout around a lattice-specific measurement.

Noisy stabilization degrades readout slowly, and the ancilla is the culprit. Figure 10 isolates the stabilization noise by reading every state under noiseless pulses. All six protocols start from the same freshly-cooled floor, $P(e) \in [0.079, 0.085]$ at $m = 0$, and rise gently and near-linearly as noisy rounds accumulate. The *rate* of that rise is set almost entirely by which channel is active during the SBS circuit, and the ordering is sharp: transmon decay is by far the most damaging ($P(e)$ climbs $0.079 \rightarrow 0.097$ over ten rounds), and the all-channels case tracks it almost exactly ($0.079 \rightarrow 0.098$), while oscillator dephasing ($0.079 \rightarrow 0.086$), oscillator decay ($0.079 \rightarrow 0.083$) and transmon dephasing ($0.079 \rightarrow 0.083$) each add only a few $\times 10^{-3}$ over the whole sweep. That the ancilla’s T_1 dominates — and that isolated oscillator loss is nearly harmless — is exactly what one expects of an actively-pumped stabilizer: the transmon is entangled with the oscillator throughout each conditional displacement, so its decay corrupts the round it drives, whereas slow photon loss is continually undone by the very map that injects it. Across the entire sweep the six protocols stay bunched within a $\lesssim 5 \times 10^{-3}$ band: once the codeword is being actively stabilized, the choice of readout dressing matters far less than how many noisy rounds it has seen and through which channel.

Readout noise adds a separable offset, and the shortest circuit wins. Figure 11 holds the stabilization at its realistic all-channels setting and sweeps the readout circuit over its six noise conditions. Adding readout noise leaves the shape of the m -dependence intact and shifts each protocol up by a nearly m -independent offset, so the two noise axes are separable. But the offset is strongly *protocol*-dependent, and it tracks the *duration* of the readout circuit rather than its correction content: single GCR — the shortest deterministic sequence, one pre-correction and one kick — is almost immune, its $P(e)$ rising by only $+2.5 \times 10^{-3}$ under all-channels readout, while the five longer circuits are lifted *together* by $+1.8$ to $+2.0 \times 10^{-2}$, whether or not they carry detuning corrections (bare BB1, with none, is lifted as much as the detuning-heavy schemes). Readout noise therefore does not blur the family uniformly; it drives single GCR clearly below the pack and widens the protocol band from $\sim 6 \times 10^{-3}$ under noiseless pulses to $\sim 2 \times 10^{-2}$. The offset again follows the transmon — all-channels and transmon-decay readout dominate (≈ 1.6 and 0.9×10^{-2} on average), oscillator decay contributes least ($\lesssim 3 \times 10^{-3}$) — reproducing the verdict of Sec. 8: under readout-circuit noise the least-exposed, shortest protocol is the most accurate, and the elaborate corrections pay for their length.

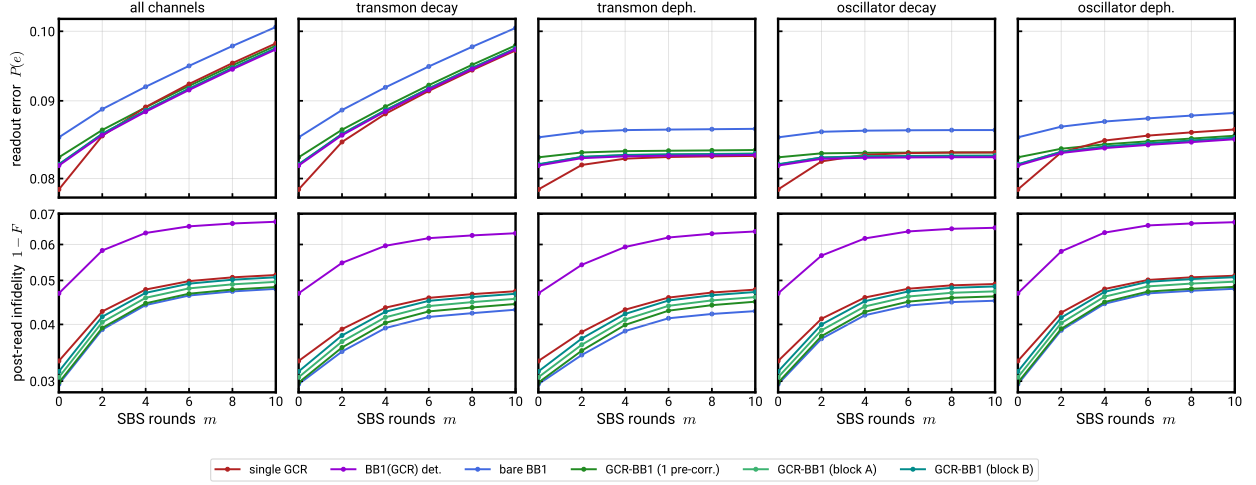


Figure 10: Reading an actively-stabilized logical 0 after m rounds of *noisy* SBS, under *noiseless* readout (isolating the stabilization noise). Columns are the five SBS-noise conditions; top row is the readout error $P(e)$, bottom row the post-read oscillator infidelity $1 - F$; each panel overlays the six deterministic protocols versus the number of rounds m (the heralded scheme is omitted, see text). Note the logarithmic vertical scale (chosen to expose the small protocol-to-protocol differences on the high binomial-SBS floor). All protocols share the freshly-cooled floor at $m = 0$ and rise gently; the rise is dominated by transmon decay (and the all-channels case tracks it), while the oscillator channels and transmon dephasing are far gentler. On $P(e)$ (top) the six protocols stay bunched within a few $\times 10^{-3}$ throughout; on $1 - F$ (bottom) BB1(GCR) det. (purple) separates cleanly as the least-faithful, sitting $\approx 1.4\text{--}1.6 \times 10^{-2}$ above the tightly-grouped pack (bare BB1, GCR-BB1 and the two blocks lowest, single GCR just above them).

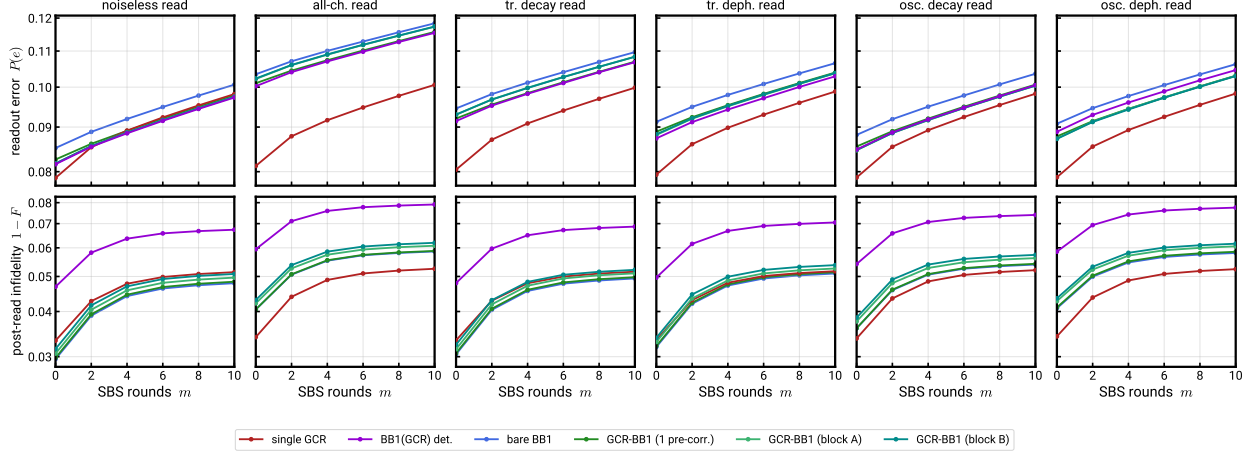


Figure 11: The same experiment with the SBS stabilization fixed at its realistic all-channels setting, now sweeping the *readout*-circuit noise across its six conditions (columns). Rows and curves as in Fig. 10; logarithmic vertical scale. Readout noise shifts the m -dependence up by a nearly constant, m -independent offset — the two noise axes are separable — largest for transmon decay and the all-channels case, smallest for oscillator decay. The offset tracks circuit *duration*: under all-channels readout the freshly-cooled ($m=0$) $P(e)$ rises by only $+2.5 \times 10^{-3}$ for single GCR (red) but by $+1.8$ to $+2.0 \times 10^{-2}$ for the five longer protocols, so single GCR pulls clearly below the pack (which lifts together regardless of correction content) — reproducing the short-circuit-wins verdict of Sec. 8.

The state’s-eye view, and the bottom line. The oscillator infidelity (bottom rows of Figs. 10–11) tells the same story with one addition. Starting from $1 - F \in [0.030, 0.047]$ at $m = 0$, the post-read infidelity rises and then *saturates* beyond $m \approx 4$: the noisy stabilizer reaches a fixed point at which entropy injected per round balances entropy pumped out, so the state stops degrading rather than drifting without bound. Here the protocol ordering does separate, and it *inverts* between the two metrics: bare BB1, which carries the *largest* readout error of the six, nonetheless leaves the *most* faithful oscillator state ($1 - F$: $0.030 \rightarrow 0.048$ over $m = 0 \rightarrow 10$), while the detuning-heavy BB1(GCR) correction — among the best at pinning the readout bit — leaves the *least* faithful state ($0.047 \rightarrow 0.067$), a clear $\approx 1.4\text{--}1.6 \times 10^{-2}$ above the rest of the pack (single GCR $0.033 \rightarrow 0.051$; GCR-BB1 and the two blocks bunched between bare BB1 and single GCR). The sharpest single comparison is single GCR versus BB1(GCR), the two protocols that best pin the bit: on $P(e)$ they are *tied* to within $\sim 10^{-3}$ ($0.079 \rightarrow 0.098$ vs $0.082 \rightarrow 0.097$ under noiseless readout), but on the oscillator state single GCR is better by $\approx 1.4 \times 10^{-2}$ at every m . So on this soft, actively-stabilized codeword BB1(GCR) *strictly loses* to single GCR — its three extra detuning corrections and three extra kicks buy nothing measurable on the bit while imprinting $\sim 1.4 \times 10^{-2}$ more back-action on the oscillator; the sub- 10^{-3} coherent-floor edge that justifies BB1(GCR) on a pristine comb (Sec. 7) is simply below the ~ 0.08 binomial floor here. As in Sec. 8, dressing the readout to sharpen the logical bit comes at the cost of disturbing the oscillator it reads. Taken together, the section’s message is reassuring for the deterministic protocols: under realistic, fully-noisy stabilization the readout error grows by only $\sim 2 \times 10^{-2}$ over ten rounds and the oscillator state saturates rather than collapses. The qualitative lessons of the earlier sections thus carry over intact to a codeword that is being actively kept alive: readout-circuit noise, not the readout construction, sets the achievable error, and the shortest state-relative circuit is the most robust to it. What does *not* carry over is any readout built around a lattice-pinned measurement — the heralded scheme, most accurate of

all on pristine Gaussian-comb codewords, cannot be pointed at the stabilized binomial codeword at all.

10 Summary

The ideal BB1(GCR) readout is genuinely excellent, but its advantage lives entirely in a *non-unitary* pre-correction $M = e^{c\hat{p}\sigma\phi}$. Deterministically no unitary sequence beats bare BB1: the washed-out unitary correction, pulse-splitting, and amplitude re-optimization all fail structurally (Table 1). The fix is the **flagged (post-selected) readout**: realize M probabilistically by heralding the *three* BB1 corrections, keeping the target a deterministic GCR. On the $\sim 17\%$ of shots where all three heralds succeed the sequence is the full BB1(GCR); these are kept, the rest discarded (a failed herald corrupts the shared readout qubit, so its direct bit is wrong, and taking all shots washes out to ~ 0.26). The heralding’s payoff is *not* the no-error point — there a single deterministic GCR already matches it at 5.7×10^{-4} — but *peak-location robustness across the correctable region*, where heralded BB1(GCR) stays up to $\sim 14\times$ flatter than a single GCR (Fig. 2b), and a *gentler* back-action than bare BB1 (Fig. 2c). On failure one cannot restore the sharp read: the original (unknown) data state cannot be re-prepared, and reusing the disturbed oscillator preserves only the coarse bit ($\langle \hat{x} \rangle$ is unchanged) — the precision degrades with each attempt, since a failed attempt’s back-action is destructive. (Heralding all four corrections instead reaches the ideal 4×10^{-7} but at only $\sim 7\%$ yield; since the heralds act on ancillas, not the readout qubit, this is still a valid heralded readout, just lower-yield than the three-herald choice.) The chief costs are hardware and yield: eight conditional displacements plus three ancilla heralds, the strong 2π correction requiring an exact block-encoding rather than the single first-order gadget, and ~ 6 attempts per accepted shot — so it is attractive only where the correctable-region robustness is worth that overhead and an end-of-line, post-selected readout fits the protocol.

Two further results place this in context. First, for *known*, re-preparable states — which covers most of a fault-tolerant workload (logical $|0\rangle / |+\rangle$ and ancilla preparation, stabilizer extraction, measure-and-reset), not only offline calibration — the no-cloning obstruction lifts, and the ideal correction is realized *deterministically in expectation* by repeat-until-success: re-prepare and re-run the shallow ($\sim 6\mu\text{s}$) heralded pulse until the flag fires (~ 6 attempts). A gate-level Kraus simulation puts one accepted shot at $\approx 3\%$ readout infidelity (transmon decoherence dominant; ancilla errors are herald-protected), and the herald pass rate is essentially noise-independent, so the retry count is stable. The costly deterministic alternative, amplitude amplification, is ruled out: its input reflections stack $\sim 3\times$ a full state preparation into one $\sim 130\mu\text{s}$ coherent circuit and lose to decoherence (≈ 0.98 infidelity). Second, a variationally compiled *deterministic* readout (Sec. 6) reaches 1.5×10^{-3} and then plateaus with depth — confirming, from the deterministic side, that no short unitary matches the heralded floor. The decisive practical point, once gate *durations* are counted (a CD of amplitude $|\beta|$ takes $|\beta|\mu\text{s}$, so the cost is the large-amplitude position kicks, not the tiny corrections): the full BB1(GCR) ($\sim 6\mu\text{s}$, dominated by the 2π kick it *shares with bare BB1*) accrues $\sim 3\%$ decoherence, which masks its coherent-floor advantage — bare BB1 and BB1(GCR) both land at $\approx 3\%$, so the GCR correction, though nearly free in time, buys nothing net under noise. Single GCR, which omits the 2π kick, is a $\sim 0.7\mu\text{s}$ circuit with the *same* coherent floor and only $\sim 0.2\%$ noise, so at $\approx 0.4\%$ it beats the full BB1(GCR) by $\sim 8\times$. Thus for a noisy end-of-line data readout the practical choice is **single GCR**; BB1(GCR)’s machinery pays off only once per-shot decoherence drops below the coherent gap $\sim 10^{-3}$, and its enduring roles are as the certification tool for the deterministic readout’s floor and the high-fidelity primitive for known-state preparation.

A full master-equation simulation (Sec. 7, $N_{\text{cav}} = 400$) confirms this ordering and settles the oscillator-back-action question. The read’s disturbance is a *state-agnostic* displacement — undone by a *single* corrective displacement $D(i\sqrt{\pi/2}/2)$ calibrated once on the no-error logical-0 and applied to every input — which restores the oscillator from $F \approx 0.07$ to $F \approx 0.97$ for single GCR (again the best under noise, since its short circuit decoheres least), leaving only an irreducible $\sim 5\%$ non-Gaussian residual. So the oscillator is preserved for reuse by that one calibrated displacement, not by stabilization; and heralding, though it provides error *detection*, does not improve the readout accuracy under noise.

Finally, two stress tests confirm that this verdict is robust to the state actually being read. Reading a codeword that has *idled* under oscillator decay or dephasing before the read (Sec. 8), and reading one that is being *actively stabilized* by noisy SBS rounds (Sec. 9), both leave the ranking intact: readout-circuit noise — dominated by the transmon — sets the achievable error, so the shortest state-relative circuit (single GCR) is the most accurate, and the elaborate corrections pay for their length. Active stabilization degrades the read only slowly (the error grows by $\sim 2 \times 10^{-2}$ over ten noisy rounds and the oscillator state saturates rather than collapses), so the deterministic protocols remain usable on a live logical memory. The one construction that does *not* survive the encoding change is the lattice-pinned heralded filter, which cannot be pointed at the actively-stabilized binomial codeword at all — a concrete cost of building readout around a measurement tied to a specific lattice.